All D-Flow / QGF experiments: evidence as of 2026-11-06T20:04:20.573772+00:00

This is a fixed as-of evidence snapshot across the complete saved experiment history. Completed, failed, partial, archived and not-yet-run studies are labeled separately. It is not a claim that the ongoing campaign has finished.

Raw QGF is a direct comparator, alongside averaged-Q and Jacobian-smoothed QGF where run. Methods share the stated frozen BC and Q within each experiment. Each method/task/trained seed uses its own validation-selected settings; fixed kappa 0.02/0.04 controls remain separate. Paired starts and display aliases do not create additional independent training seeds.

All source-table rows and all selected parameter rows are exported in the accompanying CSVs, with original file, SHA and source line. The PDF shows study summaries and source-order parameter excerpts; no omitted excerpt is treated as a missing scientific result.

Section	Study type	Status
Initial synthetic 20-task comparison	synthetic_points	COMPLETE
Synthetic full independent-lambda sweep	synthetic_points	COMPLETE
Original robot BC/IQL and positive-iteration DF pilot	native_robot	COMPLETE
Robot terminal-support and capacity diagnosis	robot_critic_diagnostic	COMPLETE
Early robot FULL/action-hole/state-hole control	native_robot	COMPLETE
Robot temporal source-noise controls	native_robot	COMPLETE
Robot IPG/FK extensions and raw-time repair	native_robot_and_sampler_diagnostic	COMPLETE
Synthetic IPG/FK and convergence/reference panels	synthetic_populations	COMPLETE
Custom sequential learned-BC/IQL navigation	custom_navigation	COMPLETE
Custom early/late-clock coverage control	custom_navigation	COMPLETE
Native PointMaze baseline	native_pointmaze	COMPLETE
Early 8,000-update critic QGF control	native_pointmaze	COMPLETE
Matched actor, increased IQL critic budget	native_pointmaze	COMPLETE
Native learned-Q action competence control	native_pointmaze	COMPLETE
First complete native D-Flow/QGF sampler comparison	native_pointmaze	COMPLETE
Native goal variation: tasks 2 and 5	native_pointmaze	COMPLETE
Native critic-training hole: failed exposure stress test	native_pointmaze	COMPLETE
Local-H first feasibility rejection	native_pointmaze	ARCHIVED
Local-H2 exposure follow-up	native_pointmaze	COMPLETE
First native endpoint-proxy geometry diagnosis	synthetic_sampling_or_saved_diagnostic	COMPLETE
Saved twin-head disagreement diagnostic	synthetic_sampling_or_saved_diagnostic	COMPLETE
Paired descriptive native outcome analysis	synthetic_sampling_or_saved_diagnostic	COMPLETE
Residual-weighted IPG, CNF score and target controls	synthetic_sampling_or_saved_diagnostic	COMPLETE
Fixed-target ULA/MH analytic panel	synthetic_sampling_or_saved_diagnostic	COMPLETE
Twenty-task ULA/MH independent sweep	synthetic_sampling_or_saved_diagnostic	COMPLETE
Privileged robot actor repair and pose follow-up	native_robot	COMPLETE
Matched H1/H5 robot action-chunk study	native_robot	COMPLETE
Proper 25D H5 IQL and terminal-invariance repair	native_robot	COMPLETE
Known narrow/broad false-Q versus genuine optima	synthetic_points	COMPLETE
Matched-target quadrature and finite-chain diagnosis	synthetic_chains	COMPLETE
Second-night all-12-family FULL/HOLE coverage comparison	native_pointmaze_modified_reset	COMPLETE
Second-night actor Jacobian and clipping diagnostic	saved_actor_geometry	COMPLETE
Current standard task1 continuation	native_pointmaze	COMPLETE
Current standard task5 completed seed11 validation	native_pointmaze	PARTIAL
Current Euler/lambda wiring and geometry results	offline_validation_diagnostic	COMPLETE
Precision-only discriminator	offline_validation_diagnostic	COMPLETE
New Q-averaged stochastic identity/full-flow prototype	engineering_only	PARTIAL

Current standard task 1: every active result

Three trained seeds are shown separately. These are the preserved original 20 paired starts per method/model, not 40. Raw QGF and the fixed 0.02/0.04 controls remain explicit. All active choices were frozen before tests.

Active requested policy	Seed 11	Seed 23	Seed 37
BC	0/20	0/20	0/20
Uniform BC N8_T8	0/20	0/20	0/20
Uniform BC N8_T32	0/20	0/20	0/20
Uniform BC N16_T8	0/20	0/20	0/20
Uniform BC N16_T32	0/20	0/20	0/20
QGF fixed 0.02	0/20	0/20	0/20
QGF fixed 0.04	0/20	0/20	0/20
Greedy Q (81 actions)	10/20	20/20	20/20
Raw QGF	13/20	20/20	20/20
DF R full/full L0	14/20	20/20	20/20
DF R full/full L1	16/20	20/20	20/20
DF R identity/full L0	12/20	20/20	20/20
DF R identity/full L1	15/20	20/20	20/20
DF S full/full L0	12/20	20/20	20/20
DF S full/full L1	12/20	20/20	20/20
DF S identity/full L0	10/20	20/20	20/20
DF S identity/full L1	10/20	20/20	20/20
DF avgQ R full/full L0	13/20	20/20	20/20
DF avgQ R full/full L1	13/20	20/20	20/20
DF avgQ S full/full L0	12/20	20/20	20/20
DF avgQ S full/full L1	11/20	20/20	20/20
DF MH full/full	11/20	20/20	20/20
DF avgQ MH full/full	11/20	20/20	20/20
QGF avg-Q	13/20	20/20	20/20
QGF smooth-J	13/20	20/20	20/20
IPG	9/20	20/20	20/20
IPG-RW	14/20	20/20	19/20

There are no exact-ID aliases or positive secondaries among these 27 active primary policies. They represent 540 actual episodes per trained seed, 1,620 in total. The separate historical ledger additionally retains 500 Euler and 80 CV/FK episodes, for 2,200 historical actual episodes. This study-specific accounting is not a sum over the overlapping report tables.

Current task 1: complete active parameter table

Requested policy	Seed 11: own setting	Seed 23: own setting	Seed 37: own setting
BC	Canonical BC (Q unused)	Canonical BC (Q unused)	Canonical BC (Q unused)
Uniform BC N8_T8	BC N8 M8	BC N8 M8	BC N8 M8
Uniform BC N8_T32	BC N8 M32	BC N8 M32	BC N8 M32
Uniform BC N16_T8	BC N16 M8	BC N16 M8	BC N16 M8
Uniform BC N16_T32	BC N16 M32	BC N16 M32	BC N16 M32
QGF fixed 0.02	k0.02	k0.02	k0.02
QGF fixed 0.04	k0.04	k0.04	k0.04
Greedy Q (81 actions)	81-action Q argmax; k unused	81-action Q argmax; k unused	81-action Q argmax; k unused
Raw QGF	k80	k80	k40
DF R full/full L0	k5 e0.1 I256 L0	k20 e0.1 I256 L0	k5 e0.1 I256 L0
DF R full/full L1	k5 e0.03 I256 L1	k80 e0.01 I128 L1	k80 e0.03 I256 L1
DF R identity/full L0	k5 e0.1 I256 L0	k40 e0.1 I256 L0	k5 e0.1 I256 L0
DF R identity/full L1	k5 e0.1 I128 L1	k80 e0.03 I256 L1	k80 e0.03 I256 L1
DF S full/full L0	k80 e0.005 I256 L0	k40 e0.1 I128 L0	k80 e0.03 I128 L0
DF S full/full L1	k80 e0.005 I256 L1	k80 e0.01 I128 L1	k80 e0.01 I128 L1
DF S identity/full L0	k80 e0.01 I128 L0	k80 e0.03 I256 L0	k80 e0.1 I256 L0
DF S identity/full L1	k80 e0.01 I128 L1	k80 e0.03 I128 L1	k80 e0.03 I128 L1
DF avgQ R full/full L0	k40 e0.1 I256 L0 s0.2 K4	k80 e0.03 I256 L0 s0.2 K4	k20 e0.1 I256 L0 s0.2 K4
DF avgQ R full/full L1	k80 e0.03 I64 L1 s0.03 K4	k80 e0.01 I256 L1 s0.01 K16	k80 e0.01 I256 L1 s0.2 K4
DF avgQ S full/full L0	k80 e0.03 I256 L0 s0.2 K4	k80 e0.03 I128 L0 s0.1 K16	k80 e0.03 I128 L0 s0.01 K4
DF avgQ S full/full L1	k80 e0.03 I128 L1 s0.01 K16	k80 e0.01 I128 L1 s0.01 K16	k80 e0.01 I128 L1 s0.1 K4
DF MH full/full	k80 e0.1 I256 L1	k80 e0.1 I32 L1	k80 e0.1 I128 L1
DF avgQ MH full/full	k80 e0.1 I256 L1 s0.01 K16	k80 e0.1 I32 L1 s0.1 K16	k80 e0.03 I128 L1 s0.1 K16
QGF avg-Q	k80 s0.1 K16	k80 s0.01 K4	k80 s0.1 K16
QGF smooth-J	k40 s0.1 K4	k40 s0.2 K16	k80 s0.1 K16
IPG	k80 N8 M32 n0.005 r0.1	k40 N16 M32 n0.02 r0.01	k40 N16 M8 n0.02 r0.01
IPG-RW	k5 N16 M8 n0.005 r0.01	k40 N16 M32 n0.02 r0.1	k40 N16 M32 n0.02 r0.1

k=kappa; e=eta; I=outer source iterations; L=lambda; s=sigma; K=action-Q panel size; N=particles; M=flow slices; n=particle noise; r=RKHS ridge. QGF eta/lambda/I are unused; particle baselines also do not use them. Greedy Q uses the critic despite unused raw kappa=0 metadata. Compact decimal display is backed by exact original values/config IDs in the selected-parameter CSV and source JSON.

Current task 1: archived Euler evidence

These methods were completed before the user removed future D-Flow Euler-proxy work. Their outcomes, own selected settings and physical costs remain preserved. They are not future pending cells.

Archived requested policy	Seed 11	Seed 23	Seed 37
DF R full/euler L0	0/20	0/20*	0/20*
DF R full/euler L1	0/20	0/20	0/20*
DF R identity/euler L0	13/20	20/20	20/20
DF R identity/euler L1	13/20	20/20	20/20
DF S full/euler L0	15/20	18/20	18/20
DF S full/euler L1	0/20	1/20	0/20
DF S identity/euler L0	12/20	20/20	20/20
DF S identity/euler L1	12/20	20/20	20/20

Archived own setting	Seed 11	Seed 23	Seed 37
DF R full/euler L0	k1 e0.1 I256 L0	Canonical BC (Q unused)	Canonical BC (Q unused)
DF R full/euler L1	k5 e0.1 I4 L1	k0 e0.01 I32 L1	Canonical BC (Q unused)
DF R identity/euler L0	k20 e0.1 I256 L0	k20 e0.1 I256 L0	k5 e0.1 I256 L0
DF R identity/euler L1	k80 e0.03 I128 L1	k80 e0.03 I256 L1	k80 e0.03 I256 L1
DF S full/euler L0	k5 e0.1 I256 L0	k5 e0.1 I256 L0	k12 e0.1 I256 L0
DF S full/euler L1	k80 e0.1 I128 L1	k80 e0.1 I256 L1	k40 e0.1 I128 L1
DF S identity/euler L0	k80 e0.01 I128 L0	k80 e0.03 I256 L0	k80 e0.03 I128 L0
DF S identity/euler L1	k80 e0.01 I128 L1	k80 e0.03 I128 L1	k80 e0.03 I128 L1

A star marks a saved display alias. Positive Euler secondary rows are retained in CSVs and original source tables. Historical IPG-CV/FK records are preserved separately in the dated reports and source archive; absence from this current-scope table is not a zero.

What the task, Q and BC mean

Initial synthetic trials score individual 2D points against target circles. Later one-dimensional analytic action/chain studies use target intervals and are labeled separately. These tests have no next state or Bellman-trained IQL critic unless explicitly stated. A point or chain count is not a robot episode.

Sequential PointMaze navigation uses real transitions and learned BC flow plus task-specific IQL Q/V. Standard task 1 goes from nominal (0,0) to (20,20), task 5 from (20,4) to (0,0), under the native 1,000-decision limit. The separate upstream-reset coverage stress uses a modified reset and 128-step cap. These cannot be pooled into one benchmark.

MuJoCo cube manipulation evaluates robot episodes and requires behavior competence, successful data and a useful critic. A weak grasp/BC baseline or critic extrapolation can limit guidance. Failed manipulation results alone do not identify one of these as the unique cause.

A learned offline Q can be wrong outside data support. Stochastic movement, Q averaging, Jacobian smoothing, particles and MH can change search or score sensitivity; they do not identify true unseen-action Q or repair its training data automatically.

The native PointMaze BC actor is goal-agnostic in its x,y input. The custom-navigation actor instead receives position, goal and clock. IQL trains the stated Q/V, while the separately frozen flow is ordinary BC rather than an IQL advantage-weighted actor. Low goal success by BC alone is not a distribution-fit test.

In learned-IQL studies, the one-step target is y=r+gamma*mask*V(s_next). Both Q heads regress toward y with squared error. V uses expectile regression toward min_target_head Q(s,logged_action). Guidance evaluates min_target_head Q(s,proposed_action), not V and not an explicit advantage. The BC flow uses ordinary flow matching independently of IQL advantage weights.

H1 uses an actual one-step successor. The proper H5 critic uses the discounted five-action reward prefix and actual successor after the prefix: y=sum_j gamma^j r_j + gamma^H mask V(s_next_H). Earlier capacity or terminal probes and actor-only robot controls are labeled separately; they are not all IQL-guided policy experiments.

Native training config and exact inference grids

These settings apply to the learned native PointMaze branch, not universally to robots or custom navigation. A 1,000-update BC actor and 50,000-update task-specific IQL critic are frozen during sampler continuation.

Native training component	Setting
Actor	Two 64-unit layers; GELU
Q / V	Two 128-unit layers; LayerNorm; two Q heads, minimum aggregation
IQL target / value	gamma=0.99; target EMA tau=0.005; expectile=0.8
Optimization	Batch 256; BC/Q/V learning rates 3e-4
Action / BC integration	H=1; eight BC flow slices

Declared sampler domain	Values
kappa	0, 0.02, 0.04, 0.06, 0.08, 0.1, 0.2, 0.3, 0.4, 0.6, 0.8, 1, 2, 3, 5, 8, 12, 20, 40, 80
eta	0.005, 0.01, 0.03, 0.1
Outer source I	1, 2, 4, 8, 16, 32, 64, 128, 256
Q averaging sigma / K	[0.01, 0.03, 0.1, 0.2] / [4, 16]
Particles N / flow slices M	[8, 16] / [8, 32]
Prior / temperature	D-Flow lambda 0/1; primary stochastic T=1; MH lambda=T=1

Each method/model/task selects independently using 16 offline validation states for the declared prescreen and 8 native validation starts for physical selection. Raw QGF evaluates all 20 kappa values. D-Flow/particles use declared prescreens and shortlists; Q averaging uses a hierarchical sigma/K search rather than the full Cartesian product. All three models freeze their choices before task tests. These are the best declared candidates, not globally optimal settings, and inference compute is unequal.

The verbatim native config and candidate-domain DESIGN are included in the evidence ZIP. Every historical training/config JSON and source-specific selection remains separately attributed; validation cost is not silently equated with training updates or physical testing.

Coefficients and the changing method scope

Symbol	Meaning	Where it acts
kappa	Q guidance multiplier	QGF vector field; D-Flow source drift; particle guidance
eta	Source step size	D-Flow drift eta and movement noise sqrt(2 eta T)
lambda	Source prior pull	-eta*lambda*z; origin, not the task goal
I	Outer source iterations	Repeated source updates before one final action
M / flow slices	BC ODE integration	Native F8 uses eight Euler integration steps
sigma / K	Q-score smoothing / panel size	Nearby final-action Q average
T	Movement temperature	Primary stochastic T=1; independent of sigma

D-Flow: z_next=z+eta[kappa*g(z)-lambda*z]+sqrt(2*eta*T)*eps. Regular variants omit movement noise. Kappa and eta have different roles; the same numeric kappa is not equal effective guidance across different methods or critics.

Full-J/full guidance differentiates Q(F(z)); the learned native F is the numerically integrated, clipped F8. Historical analytic toys instead use their declared analytic full map. Historical full-J/Euler guidance differentiates a one-Euler endpoint proxy, then still executes the full final map. Identity guidance bypasses that endpoint Jacobian.

For bounded native actions, QGF refreshes velocity, endpoint and action-Q gradient every pseudo-time slice: E_t=clip[a_t+(1-t)v(s,a_t,t)]; a_next=a_t+dt[v(s,a_t,t)+kappa*grad_a Q(s,E_t)], dt=1/N. Native learned experiments use N=8. Original analytic toys omit endpoint clipping and declare N=32. QGF has no eta/lambda/I outer source loop.

Qbar(s,a)=mean_k min_head Q_head(s,clip[a+sigma*eps_k]). A finite antithetic panel stays fixed within one action call, including both MH sides. Averaging changes the score; source noise changes exploration. Jacobian smoothing is distinct. MH proposes and accepts, or keeps its previous source; finite mixing and critic accuracy remain unproven.

Every D-Flow family and what remains active

Family	Source update	Backward / endpoint	Future scope
1	Regular	Full J / full map	Active; lambda 0/1
2	Regular	Identity / full map	Active; lambda 0/1
3	Regular	Full J / Euler proxy	Archived; lambda 0/1
4	Regular	Identity / Euler proxy	Archived; lambda 0/1
5	Stochastic	Full J / full map	Active; lambda 0/1
6	Stochastic	Identity / full map	Active; lambda 0/1
7	Stochastic	Full J / Euler proxy	Archived; lambda 0/1
8	Stochastic	Identity / Euler proxy	Archived; lambda 0/1
9	Regular + Q average	Full J / full map	Active; lambda 0/1
10	Stochastic + Q average	Full J / full map	Active; lambda 0/1
11	Raw-Q MH	Full J / full map	Active; lambda=T=1
12	Q-averaged MH	Full J / full map	Active; lambda=T=1
13, new	Stochastic + Q average	Identity / full map	Exploratory; lambda 0/1; no native results

The original 2 x 2 x 2 product is regular/stochastic x full/identity Jacobian x full/Euler guidance endpoint. Four later families make the historical 12 families / 22 lambda conditions. Removing four Euler families leaves eight core families / 14 conditions. The new family is historically 13th and ninth active, giving 16 active D-Flow conditions when its two exploratory strata are included.

Full-flow here means using the complete BC map as the guidance endpoint. F8 still approximates its ODE with eight Euler integration steps. The removed one-step endpoint proxy, the integrator steps, outer source I and physical decisions are four distinct concepts.

Why QGF remains: it refreshes E_t and its gradient at every slice. Historical D-Flow Euler guidance repeatedly uses $E0(z)=\mathrm{clip}[z+v(s,z,0)]$ in the outer source loop, then executes F8. Under ideal population-optimal squared-error flow matching with independent Gaussian source and straight interpolation, E0 is the conditional mean action and its source Jacobian cancels. That identity depends on the assumptions; measured learned Jacobians are attenuated but nonzero. It is not proof of task failure or Q accuracy.

IPG transports particles and selects uniformly; IPG-RW retains residual importance weights and selects with those weights. This is sampler weighting, not actor/Q retraining. Historical IPG-CV and FK remain archived. New 40-start evaluation is preserved A20 plus fresh B20 using exactly frozen configurations; missing B20 is pending, not zero.

Initial synthetic 20-task comparison [COMPLETE]

Task / success: One generated 2D point must enter a task target circle; no next state or IQL V.

Q: Fitted synthetic immediate-reward RBF critic; controlled omissions/false peaks, not IQL. True reward is separately known for validation/test.

BC: Analytic 2D behavior transport, not a learned neural BC policy.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

BC succeeded in 56/400 point trials; tuned QGF in 166/400; initial regular full-J/full-flow D-Flow in 201/400. The main setting used lambda=0; lambda=1 reused those selected settings rather than being tuned independently.

Task / seed	Method	Own frozen setting
mode01 /	bc	BC control; Q guidance unused; see source variant
mode01 /	dflow_det_full_full	kappa=5.0 eta=0.03 lambda=0.0 iterations=128
mode01 /	dflow_det_identity_full	kappa=5.0 eta=0.01 lambda=0.0 iterations=128
mode01 /	dflow_det_full_euler	kappa=1.0 eta=0.1 lambda=0.0 iterations=32
mode01 /	dflow_det_identity_euler	kappa=0.6 eta=0.03 lambda=0.0 iterations=128
mode01 /	dflow_stoch_full_full	kappa=5.0 eta=0.01 lambda=0.0 iterations=32
mode01 /	dflow_stoch_identity_full	kappa=5.0 eta=0.03 lambda=0.0 iterations=64
mode01 /	dflow_stoch_full_euler	kappa=5.0 eta=0.03 lambda=0.0 iterations=32
mode01 /	dflow_stoch_identity_euler	kappa=5.0 eta=0.01 lambda=0.0 iterations=64
mode01 /	qgf	kappa=2.0
mode02 /	bc	BC control; Q guidance unused; see source variant
mode02 /	dflow_det_full_full	kappa=2.0 eta=0.1 lambda=0.0 iterations=128

Parameter excerpt: first 12 of 260 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Superseded as the main synthetic comparison by independent-lambda S02. These are conditional inference trials, not trained model seeds.

Evidence: outputs/comparison/README.txt; table/source hashes are recorded in the snapshot and file catalog.

Synthetic full independent-lambda sweep [COMPLETE]

Task / success: One generated 2D point must enter a task target circle; no next state or IQL V.

Q: Fitted synthetic immediate-reward RBF critic; controlled omissions/false peaks, not IQL. True reward is separately known for validation/test.

BC: Analytic 2D behavior transport, not a learned neural BC policy.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

Twenty tasks used 40 validation sources and 20 fresh heldout sources per task. Tuned QGF: 166/400; fixed kappa 0.02/0.04: 60/400 and 62/400; regular full-J/full-flow D-Flow with lambda=1: 209/400. All 16 historical D-Flow conditions remain visible.

Task / seed	Method	Own frozen setting
mode01 /	bc	BC control; Q guidance unused; see source variant
mode01 /	dflow_det_full_full__lambda0	kappa=5.0 eta=0.03 lambda=0.0 iterations=128
mode01 /	dflow_det_identity_full__lambda0	kappa=0.8 eta=0.03 lambda=0.0 iterations=256
mode01 /	dflow_det_full_euler__lambda0	kappa=1.0 eta=0.1 lambda=0.0 iterations=32
mode01 /	dflow_det_identity_euler__lambda0	kappa=0.6 eta=0.03 lambda=0.0 iterations=128
mode01 /	dflow_stoch_full_full__lambda0	kappa=5.0 eta=0.01 lambda=0.0 iterations=256
mode01 /	dflow_stoch_identity_full__lambda0	kappa=5.0 eta=0.03 lambda=0.0 iterations=256
mode01 /	dflow_stoch_full_euler__lambda0	kappa=5.0 eta=0.005 lambda=0.0 iterations=256
mode01 /	dflow_stoch_identity_euler__lambda0	kappa=5.0 eta=0.01 lambda=0.0 iterations=256
mode01 /	dflow_det_full_full__lambda1	kappa=5.0 eta=0.1 lambda=1.0 iterations=16
mode01 /	dflow_det_identity_full__lambda1	kappa=5.0 eta=0.005 lambda=1.0 iterations=128
mode01 /	dflow_det_full_euler__lambda1	kappa=0.3 eta=0.1 lambda=1.0 iterations=64

Parameter excerpt: first 12 of 460 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Analytic transport geometry differs from learned linear-path FM; oracle validation and unequal compute. No tuned-QGF task had zero successes.

Evidence: outputs/comparison_sweeps/PROTOCOL_NOTES.txt; outputs/comparison_sweeps/README.txt; table/source hashes are recorded in the snapshot and file catalog.

Original robot BC/IQL and positive-iteration DF pilot [COMPLETE]

Task / success: Native cube-triple robot: yellow cube to its goal while other cubes remain within their goals; all three centers within 4 cm, up to 1000 decisions. Native reward is number of correctly placed cubes minus 3; success can occur while a cube is still held.

Q: Learned IQL target-min Q and V on actual robot transitions; horizon1 has 5D actions, horizon5 critic has 25D discounted action prefixes.

BC: Learned robot BC flow; privileged successful-demo augmentation is explicitly separated from original logged data.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Original robot stage: a 1,000,000-transition TRAIN slice plus 100,000 VAL records, 50,000 BC/IQL updates, two 64-unit actor layers, two 128-unit Q/V layers, batch 256, H=1, gamma=0.999 and eight flow slices. The logged slice had no full-task success states. The coverage stage used a separate matched 200,000-transition training slice and 20,000 updates, with a shared actor/critic; it is not the same training stage.

Primary D-Flow selections fell back to BC. Separate positive-iteration rows were mostly 0/20; regular full-J/full-flow lambda=1 reached 1/20. Tuned QGF selected kappa=0; fixed 0.02/0.04 remained 0/20.

Task / seed	Method	Own frozen setting
/	dflow_det_full_full__lambda0	kappa=5.0 eta=0.1 lambda=0.0 iterations=8
/	dflow_det_identity_full__lambda0	kappa=0.1 eta=0.01 lambda=0.0 iterations=8
/	dflow_det_full_euler__lambda0	kappa=0.08 eta=0.1 lambda=0.0 iterations=1
/	dflow_det_identity_euler__lambda0	kappa=0.08 eta=0.1 lambda=0.0 iterations=1
/	dflow_det_full_full__lambda1	kappa=1.0 eta=0.01 lambda=1.0 iterations=4
/	dflow_det_identity_full__lambda1	kappa=0.4 eta=0.03 lambda=1.0 iterations=1
/	dflow_det_full_euler__lambda1	kappa=2.0 eta=0.03 lambda=1.0 iterations=1
/	dflow_det_identity_euler__lambda1	kappa=0.4 eta=0.03 lambda=1.0 iterations=1
/	dflow_det_full_full__lambda0	kappa=5.0 eta=0.1 lambda=0.0 iterations=8
/	dflow_det_identity_full__lambda0	kappa=0.1 eta=0.01 lambda=0.0 iterations=8
/	dflow_det_full_euler__lambda0	kappa=0.08 eta=0.1 lambda=0.0 iterations=1
/	dflow_det_identity_euler__lambda0	kappa=0.08 eta=0.1 lambda=0.0 iterations=1

Parameter excerpt: first 12 of 74 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Small CPU pilot; one trained model and a limited logged slice, not a full benchmark reproduction. Original/secondary rows are separate.

Evidence: outputs/official_pilot/ROBOT_REPORT.txt; outputs/official_pilot/regular_worker/PILOT_SCOPE.txt; outputs/official_pilot/stoch_worker/PILOT_SCOPE.txt; table/source hashes are recorded in the snapshot and file catalog.

Robot terminal-support and capacity diagnosis [COMPLETE]

Task / success: Native cube-triple robot: yellow cube to its goal while other cubes remain within their goals; all three centers within 4 cm, up to 1000 decisions. Native reward is number of correctly placed cubes minus 3; success can occur while a cube is still held.

Q: Matched 10,000-update critic/value continuations compare original-only logged data with additional privileged oracle-transition support. Known terminal labels are assessed separately from general critic error.

BC: Learned robot BC flow; privileged successful-demo augmentation is explicitly separated from original logged data. Shared frozen50k actor.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

After a fixed 10,000-update continuation, terminal-test RMSE was 302.45 with original data and 19.09 with augmentation. The four terminal states provide known zero boundaries, not general Q-error labels. This diagnostic ran no behavior-policy episodes.

Limit: 12 oracle training episodes,4validation and4terminal tests are privileged evidence. Capacity probe is distinct from IQL and has retention/overfit costs.

Evidence: outputs/critic_repair/REPAIR_SCOPE.txt; outputs/critic_repair/action_ranking/READ_ME.txt; table/source hashes are recorded in the snapshot and file catalog.

Early robot FULL/action-hole/state-hole control [COMPLETE]

Task / success: Native cube-triple robot: yellow cube to its goal while other cubes remain within their goals; all three centers within 4 cm, up to 1000 decisions. Native reward is number of correctly placed cubes minus 3; success can occur while a cube is still held.

Q: Learned IQL target-min Q and V on actual robot transitions; horizon1 has 5D actions, horizon5 critic has 25D discounted action prefixes.

BC: Shared full-coverage200k-transition/20k-update actor.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

Every reported 20-episode method/coverage condition had zero success. Actual hole-visitation fractions remain in the source tables.

Task / seed	Method	Own frozen setting
/	bc	BC control; Q guidance unused; see source variant
/	qgf_kappa0.02	kappa=0.02
/	qgf_kappa0.04	kappa=0.04
/	dflow_det_full_full__lambda0	kappa=0.4 eta=0.03 lambda=0.0 iterations=1
/	dflow_stoch_identity_full__lambda0	kappa=5.0 eta=0.03 lambda=0.0 iterations=1
/	dflow_det_full_full__lambda1	kappa=5.0 eta=0.01 lambda=1.0 iterations=4
/	dflow_stoch_identity_full__lambda1	kappa=5.0 eta=0.03 lambda=1.0 iterations=1
/	bc	BC control; Q guidance unused; see source variant
/	qgf_kappa0.02	kappa=0.02
/	qgf_kappa0.04	kappa=0.04
/	dflow_det_full_full__lambda0	kappa=0.4 eta=0.03 lambda=0.0 iterations=1
/	dflow_stoch_identity_full__lambda0	kappa=5.0 eta=0.03 lambda=0.0 iterations=1

Parameter excerpt: first 12 of 42 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Weak task competence prevents a sampler ranking; learned Q changes and hole visits are not ground-truth extrapolation-error measurements.

Evidence: ; table/source hashes are recorded in the snapshot and file catalog.

Robot temporal source-noise controls [COMPLETE]

Task / success: Native cube-triple robot: yellow cube to its goal while other cubes remain within their goals; all three centers within 4 cm, up to 1000 decisions. Native reward is number of correctly placed cubes minus 3; success can occur while a cube is still held.

Q: Frozen robot Q/V were unused: these temporal BC controls had kappa=0 and zero critic-gradient evaluations.

BC: The frozen original logged-data 50,000-update pilot BC actor, without privileged demonstration repair or critic guidance.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

Holding one BC source for 5 steps, or for a whole episode, each scored 0/20. These are zero-guidance temporal controls.

Task / seed	Method	Own frozen setting
/	bc_hold_source_5_steps	BC control; Q guidance unused; see source variant
/	bc_fixed_source_episode	BC control; Q guidance unused; see source variant

Parameter excerpt: first 2 of 2 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Tests temporal coherence only; no demonstrated grasp or critic repair.

Evidence: ; table/source hashes are recorded in the snapshot and file catalog.

Robot IPG/FK extensions and raw-time repair [COMPLETE]

Task / success: Native cube-triple robot: yellow cube to its goal while other cubes remain within their goals; all three centers within 4 cm, up to 1000 decisions. Native reward is number of correctly placed cubes minus 3; success can occur while a cube is still held.

Q: Learned IQL target-min Q and V on actual robot transitions; horizon1 has 5D actions, horizon5 critic has 25D discounted action prefixes.

BC: Learned robot BC flow; privileged successful-demo augmentation is explicitly separated from original logged data.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

Original, ordinary-IQL-continuation and oracle-augmented critics were tested separately. Primary zero-guidance controls failed; positive IPG and IPG-CV each achieved 1/20 on the same paired reset. Diffusion and FK failures remain visible.

Task / seed	Method	Own frozen setting
/	bc	BC control; Q guidance unused; see source variant
/	qgf_kappa0.02	kappa=0.02
/	qgf_kappa0.04	kappa=0.04
/	uniform_particle_bc_N8	kappa=0.0 flow_steps=8 particles=8 noise_strength=0.0 ridge=0.1
/	uniform_particle_bc_N16	kappa=0.0 flow_steps=8 particles=16 noise_strength=0.0 ridge=0.1
/	ipg	kappa=0.0 particles=8 flow_steps=8 noise_strength=0.0 ridge=0.1
/	ipg_cv	kappa=0.0 particles=8 flow_steps=8 noise_strength=0.0 ridge=0.1
/	fkc	kappa=0.0 particles=8 flow_steps=8 noise_strength=0.0
/	ipg__positive_kappa	kappa=0.1 flow_steps=8 particles=8 noise_strength=0.0 ridge=0.01
/	ipg_cv__positive_kappa	kappa=0.1 flow_steps=8 particles=8 noise_strength=0.0 ridge=0.01
/	fkc__positive_kappa	kappa=0.4 flow_steps=8 particles=8 noise_strength=0.0 ridge=0.1
/	bc	BC control; Q guidance unused; see source variant

Parameter excerpt: first 12 of 66 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Continuous velocity interpolation/time derivatives, bridge-score and finite-particle approximations are explicit; privileged evidence is not original-only offline RL.

Evidence: outputs/particle_extension/BUNDLE_READ_ME.txt; outputs/particle_extension/PARTICLE_EXTENSION_REPORT.txt; table/source hashes are recorded in the snapshot and file catalog.

Synthetic IPG/FK and convergence/reference panels [COMPLETE]

Task / success: One generated 2D point must enter a task target circle; no next state or IQL V.

Q: Fitted synthetic immediate-reward RBF critic; controlled omissions/false peaks, not IQL. True reward is separately known for validation/test.

BC: Analytic 2D behavior transport, not a learned neural BC policy.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

IPG achieved 57/400 and FK 164/400 point trials; earlier QGF 166/400 is a reference. Exact-prior importance estimates exposed finite-population target mismatch.

Task / seed	Method	Own frozen setting
mode01 /	ipg_diagnostic_fixed_settings	kappa=5.0 flow_steps=32 particles=16 noise_strength=0.005 ridge=0.1
mode01 /	ipg_diagnostic_fixed_settings	kappa=5.0 flow_steps=32 particles=16 noise_strength=0.005 ridge=0.01
mode01 /	ipg_diagnostic_fixed_settings	kappa=5.0 flow_steps=32 particles=64 noise_strength=0.005 ridge=0.1
mode01 /	ipg_diagnostic_fixed_settings	kappa=5.0 flow_steps=32 particles=64 noise_strength=0.005 ridge=0.01
mode01 /	ipg_diagnostic_fixed_settings	kappa=5.0 flow_steps=128 particles=16 noise_strength=0.005 ridge=0.01
narrow01 /	ipg_diagnostic_fixed_settings	kappa=5.0 flow_steps=32 particles=16 noise_strength=0.005 ridge=0.1
narrow01 /	ipg_diagnostic_fixed_settings	kappa=5.0 flow_steps=32 particles=16 noise_strength=0.005 ridge=0.01
narrow01 /	ipg_diagnostic_fixed_settings	kappa=5.0 flow_steps=32 particles=64 noise_strength=0.005 ridge=0.1
narrow01 /	ipg_diagnostic_fixed_settings	kappa=5.0 flow_steps=32 particles=64 noise_strength=0.005 ridge=0.01
narrow01 /	ipg_diagnostic_fixed_settings	kappa=5.0 flow_steps=128 particles=16 noise_strength=0.005 ridge=0.01
missing01 /	ipg_diagnostic_fixed_settings	kappa=5.0 flow_steps=32 particles=16 noise_strength=0.005 ridge=0.1
missing01 /	ipg_diagnostic_fixed_settings	kappa=5.0 flow_steps=32 particles=16 noise_strength=0.005 ridge=0.01

Parameter excerpt: first 12 of 100 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Interacting particles are not independent trials. The fixed 20-configuration / 40-population diagnostic and 100,000 reference draws are separate from native episodes.

Evidence: outputs/particle_toys/README.txt; outputs/particle_toys/convergence/DIAGNOSTIC_NOTES.txt; table/source hashes are recorded in the snapshot and file catalog.

Custom sequential learned-BC/IQL navigation [COMPLETE]

Task / success: Four custom kinematic navigation MDPs: state [x,y,goal_x,goal_y,time_left], displacement 0.12*a, at most 80 decisions, success at distance <0.10. There are actual next states. This is not native OGBench or cube manipulation.

Q: Learned IQL Q/V on complete custom-MDP transition tuples.

BC: Learned custom conditional BC flow;12 task/training-seed models.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Task	Geometry / coverage
open_reach	No walls
upper_gate	One wall; pass above it
zigzag	Two walls
zigzag_action_hole	Same walls; omit complete tuples with |x|<0.16, |y|<0.45 and a_y<-0.4

Training and sampler budgets: Eight hundred behavior episodes mix a noisy planner with 20% random behavior. Training uses 8,000 updates, gamma=0.98, target EMA tau=0.005 and IQL expectile=0.8, with two 64-unit actor layers and two 128-unit Q/V layers.

There were 8,880 actual guidance episodes: 6,720 primary and 2,160 positive secondary, plus 7,488 validation episodes. BC: 221/240; tuned QGF: 230/240; regular full-J/full-flow lambda=1: 237/240.

Task / seed	Method	Own frozen setting
open_reach / 11	source record 1	See exact source row / original table
open_reach / 11	source record 2	See exact source row / original table
open_reach / 11	source record 3	See exact source row / original table
open_reach / 11	source record 4	See exact source row / original table
open_reach / 11	source record 5	See exact source row / original table
open_reach / 11	source record 6	See exact source row / original table
open_reach / 11	source record 7	See exact source row / original table
open_reach / 11	source record 8	See exact source row / original table
open_reach / 11	source record 9	See exact source row / original table
open_reach / 11	source record 10	See exact source row / original table
open_reach / 11	source record 11	See exact source row / original table
open_reach / 11	source record 12	See exact source row / original table

Parameter excerpt: first 12 of 252 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: BC is near a ceiling. Some improvements use kappa=0 source-prior shrinkage rather than Q guidance. Twelve models share paired starts.

Evidence: outputs/overnight_research/navigation_v1/NAVIGATION_REPORT.txt; table/source hashes are recorded in the snapshot and file catalog.

Custom early/late-clock coverage control [COMPLETE]

Task / success: A custom 80-decision navigation coverage follow-up changes early/late-clock transition support while retaining actual next states. It is separate from native OGBench and cube manipulation.

Q: Six 12,000-update Q/V continuations. Actual terminal labels are assessed separately from learned-V Bellman consistency.

BC: Shared fixed custom BC.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

There were 420 actual tests and 1,032 validation episodes. BC: 50/60; own-kappa parent, early-clock and late-clock conditions: 52/60, 60/60 and 58/60. Improved terminal calibration did not improve control beyond the early-data condition.

Task / seed	Method	Own frozen setting
/ 11	shared_actor	kappa=0.0
/ 11	parent	kappa=0.6
/ 11	parent	kappa=0.04
/ 11	early	kappa=0.6
/ 11	early	kappa=0.04
/ 11	late	kappa=1.0
/ 11	late	kappa=0.04
/ 23	shared_actor	kappa=0.0
/ 23	parent	kappa=0.8
/ 23	parent	kappa=0.04
/ 23	early	kappa=2.0
/ 23	early	kappa=0.04

Parameter excerpt: first 12 of 21 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Calibration improvement does not establish policy improvement; not native robot episodes.

Evidence: outputs/overnight_research/navigation_clock_coverage/CLOCK_COVERAGE_REPORT.txt; table/source hashes are recorded in the snapshot and file catalog.

Native PointMaze baseline [COMPLETE]

Task / success: Native MuJoCo PointMaze medium, repeated bounded 2D actions, actual successors and sparse task reward; 1000 native decisions unless explicitly modified.

Q: Learned IQL twin target-min Q and separate V, using actual (s,a,r,s_next,mask) tuples; exact update budgets differ by study.

BC: Learned BC flow conditioned on position x,y, with no explicit goal input. The full-data actor is frozen within a sampler comparison; task-specific Q/V provide goal preferences.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

Three trained BC actors achieved 0/60 heldout starts; the native BFS planner achieved 8/8. The 1,000-update actor checkpoint was selected by a frozen tie rule.

Limit: Cohorts differ across studies. A total out of 60 describes three trained models sharing 20 paired starts, not 60 independently trained models. Fixed/tuned settings, actual executions and aliases stay separate.

Evidence: outputs/overnight_research/pointmaze_v1/POINTMAZE_BASELINE_REPORT.txt; table/source hashes are recorded in the snapshot and file catalog.

Early 8,000-update critic QGF control [COMPLETE]

Task / success: Native MuJoCo PointMaze medium, repeated bounded 2D actions, actual successors and sparse task reward; 1000 native decisions unless explicitly modified.

Q: Learned IQL twin target-min Q and separate V, using actual (s,a,r,s_next,mask) tuples; exact update budgets differ by study.

BC: Learned BC flow conditioned on position x,y, with no explicit goal input. The full-data actor is frozen within a sampler comparison; task-specific Q/V provide goal preferences.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

Ninety validation settings used 720 episodes, followed by 400 actual tests. All success counts were zero despite route progress.

Task / seed	Method	Own frozen setting
/ 11	bc	BC control; Q guidance unused; see source variant
/ 11	qgf_kappa0.02	kappa=0.02
/ 11	qgf_kappa0.04	kappa=0.04
/ 11	qgf_tuned	Canonical BC; Q/source-loop fields unused
/ 11	qgf_tuned__positive_kappa	kappa=0.02
/ 11	qgf_kappa0.02	kappa=0.02
/ 11	qgf_kappa0.04	kappa=0.04
/ 11	qgf_tuned	kappa=5.0
/ 23	bc	BC control; Q guidance unused; see source variant
/ 23	qgf_kappa0.02	kappa=0.02
/ 23	qgf_kappa0.04	kappa=0.04
/ 23	qgf_tuned	kappa=0.3

Parameter excerpt: first 12 of 22 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Cohorts differ across studies. A total out of 60 describes three trained models sharing 20 paired starts, not 60 independently trained models. Fixed/tuned settings, actual executions and aliases stay separate.

Evidence: outputs/overnight_research/pointmaze_qgf_v1/POINTMAZE_QGF_REPORT.txt; table/source hashes are recorded in the snapshot and file catalog.

Matched actor, increased IQL critic budget [COMPLETE]

Task / success: Native MuJoCo PointMaze medium, repeated bounded 2D actions, actual successors and sparse task reward; 1000 native decisions unless explicitly modified.

Q: Learned IQL twin target-min Q and separate V, using actual (s,a,r,s_next,mask) tuples; exact update budgets differ by study.

BC: Learned BC flow conditioned on position x,y, with no explicit goal input. The full-data actor is frozen within a sampler comparison; task-specific Q/V provide goal preferences.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

With the same 1,000-update actor and the same data, tuned QGF with an 8,000-update critic achieved 0/60; with a 50,000-update critic it achieved 55/60 on fresh 232xxx starts. Fixed 0.02/0.04 controls remained zero. This added 126,000 Q/V updates and no actor updates across the three seeds.

Task / seed	Method	Own frozen setting
/ 11	bc	BC control; Q guidance unused; see source variant
/ 11	qgf_kappa0.02	kappa=0.02
/ 11	qgf_kappa0.04	kappa=0.04
/ 11	qgf_tuned	kappa=8.0
/ 11	qgf_kappa0.02	kappa=0.02
/ 11	qgf_kappa0.04	kappa=0.04
/ 11	qgf_tuned	kappa=20.0
/ 23	bc	BC control; Q guidance unused; see source variant
/ 23	qgf_kappa0.02	kappa=0.02
/ 23	qgf_kappa0.04	kappa=0.04
/ 23	qgf_tuned	kappa=2.0
/ 23	qgf_kappa0.02	kappa=0.02

Parameter excerpt: first 12 of 21 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Cohorts differ across studies. A total out of 60 describes three trained models sharing 20 paired starts, not 60 independently trained models. Fixed/tuned settings, actual executions and aliases stay separate.

Evidence: outputs/overnight_research/pointmaze_critic_budget_v1/POINTMAZE_CRITIC_BUDGET_REPORT.txt; table/source hashes are recorded in the snapshot and file catalog.

Native learned-Q action competence control [COMPLETE]

Task / success: Native MuJoCo PointMaze medium, repeated bounded 2D actions, actual successors and sparse task reward; 1000 native decisions unless explicitly modified.

Q: Learned IQL twin target-min Q and separate V, using actual (s,a,r,s_next,mask) tuples; exact update budgets differ by study.

BC: Learned BC flow conditioned on position x,y, with no explicit goal input. The full-data actor is frozen within a sampler comparison; task-specific Q/V provide goal preferences.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

On fresh 262xxx starts, the 50,000-update greedy critic control achieved 54/60 and transferred QGF 56/60. Both 8,000-update versions achieved 0/60. The 81-action greedy grid bypasses the actor.

Task / seed	Method	Own frozen setting
/ 11	bc	BC control; Q guidance unused; see source variant
/ 11	greedy_q_grid	81-action Q argmax; kappa/eta/lambda/l unused
/ 11	qgf_transferred	kappa=8.0
/ 11	greedy_q_grid	81-action Q argmax; kappa/eta/lambda/l unused
/ 11	qgf_transferred	kappa=20.0
/ 23	bc	BC control; Q guidance unused; see source variant
/ 23	greedy_q_grid	81-action Q argmax; kappa/eta/lambda/l unused
/ 23	qgf_transferred	kappa=2.0
/ 23	greedy_q_grid	81-action Q argmax; kappa/eta/lambda/l unused
/ 23	qgf_transferred	kappa=20.0
/ 37	bc	BC control; Q guidance unused; see source variant
/ 37	greedy_q_grid	81-action Q argmax; kappa/eta/lambda/l unused

Parameter excerpt: first 12 of 15 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Cohorts differ across studies. A total out of 60 describes three trained models sharing 20 paired starts, not 60 independently trained models. Fixed/tuned settings, actual executions and aliases stay separate.

Evidence: outputs/overnight_research/pointmaze_greedy_control_v1/POINTMAZE_GREEDY_REPORT.txt; table/source hashes are recorded in the snapshot and file catalog.

First complete native D-Flow/QGF sampler comparison [COMPLETE]

Task / success: Native MuJoCo PointMaze medium, repeated bounded 2D actions, actual successors and sparse task reward; 1000 native decisions unless explicitly modified.

Q: Learned IQL twin target-min Q and separate V, using actual (s,a,r,s_next,mask) tuples; exact update budgets differ by study.

BC: Learned BC flow conditioned on position x,y, with no explicit goal input. The full-data actor is frozen within a sampler comparison; task-specific Q/V provide goal preferences.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

All 75 choices froze before testing. There were 3,912 validation episodes and 2,040 actual tests across 102 unique configurations. BC: 0/60; QGF: 53/60; historical regular identity-J/Euler lambda=1: 58/60. Original failed Euler conditions remain in the record.

Task / seed	Method	Own frozen setting
/ 11	bc	Canonical BC; Q/source-loop fields unused
/ 11	bc_particles_N8_T8	particles=8 flow_steps=8
/ 11	bc_particles_N8_T32	particles=8 flow_steps=32
/ 11	bc_particles_N16_T8	particles=16 flow_steps=8
/ 11	bc_particles_N16_T32	particles=16 flow_steps=32
/ 11	greedy_q_grid	81-action Q argmax; kappa/eta/lambda/I unused
/ 11	qgf_fixed_0.02	kappa=0.02
/ 11	qgf_fixed_0.04	kappa=0.04
/ 11	qgf_fixed_20	kappa=20.0
/ 11	dflow_det_full_full__lambda0	kappa=5.0 eta=0.1 lambda=0.0 iterations=256
/ 11	dflow_det_full_full__lambda1	kappa=40.0 eta=0.01 lambda=1.0 iterations=256
/ 11	dflow_det_identity_full__lambda0	kappa=12.0 eta=0.03 lambda=0.0 iterations=256

Parameter excerpt: first 12 of 468 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Cohorts differ across studies. A total out of 60 describes three trained models sharing 20 paired starts, not 60 independently trained models. Fixed/tuned settings, actual executions and aliases stay separate.

Evidence: outputs/overnight_research/pointmaze_full_samplers/POINTMAZE_FULL_COMPARISON_REPORT.txt; outputs/overnight_research/pointmaze_full_samplers/kernel_preflight/README.txt; table/source hashes are recorded in the snapshot and file catalog.

Native goal variation: tasks 2 and 5 [COMPLETE]

Task / success: Native MuJoCo PointMaze medium, repeated bounded 2D actions, actual successors and sparse task reward; 1000 native decisions unless explicitly modified.

Q: Learned IQL twin target-min Q and separate V, using actual (s,a,r,s_next,mask) tuples; exact update budgets differ by study.

BC: Learned BC flow conditioned on position x,y, with no explicit goal input. The full-data actor is frozen within a sampler comparison; task-specific Q/V provide goal preferences.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

Goal-specific 50,000-update critics gave tuned QGF 60/60 on task 2 and 49/60 on task 5. Fixed kappa=20 gave 60/60 and 47/60. There were 680 actual tests, 912 validation episodes and 16 planner checks.

Task / seed	Method	Own frozen setting
2 / 11	bc	BC control; Q guidance unused; see source variant
2 / 11	qgf_fixed.02	kappa=0.02
2 / 11	qgf_fixed.04	kappa=0.04
2 / 11	qgf_fixed20	kappa=20.0
2 / 11	qgf_tuned	kappa=12.0
2 / 11	greedy_grid81	kappa=N/A
2 / 23	bc	BC control; Q guidance unused; see source variant
2 / 23	qgf_fixed.02	kappa=0.02
2 / 23	qgf_fixed.04	kappa=0.04
2 / 23	qgf_fixed20	kappa=20.0
2 / 23	qgf_tuned	kappa=40.0
2 / 23	greedy_grid81	kappa=N/A

Parameter excerpt: first 12 of 42 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Cohorts differ across studies. A total out of 60 describes three trained models sharing 20 paired starts, not 60 independently trained models. Fixed/tuned settings, actual executions and aliases stay separate.

Evidence: outputs/overnight_research/pointmaze_task_variation/TASK_VARIATION_REPORT.txt; table/source hashes are recorded in the snapshot and file catalog.

Native critic-training hole: failed exposure stress test [COMPLETE]

Task / success: Native MuJoCo PointMaze medium, repeated bounded 2D actions, actual successors and sparse task reward; 1000 native decisions unless explicitly modified.

Q: Learned IQL twin target-min Q and separate V, using actual (s,a,r,s_next,mask) tuples; exact update budgets differ by study.

BC: Learned BC flow conditioned on position x,y, with no explicit goal input. The full-data actor is frozen within a sampler comparison; task-specific Q/V provide goal preferences.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

There were 888 validation episodes and 660 actual tests. FULL QGF achieved 54/60 and HOLE QGF 52/60, but all 660 test episodes avoided region H. This did not directly test unseen-H exposure.

Task / seed	Method	Own frozen setting
/ 11	bc	BC control; Q guidance unused; see source variant
/ 23	bc	BC control; Q guidance unused; see source variant
/ 37	bc	BC control; Q guidance unused; see source variant
/ 11	primary	kappa=12.0
/ 23	primary	kappa=40.0
/ 37	primary	kappa=40.0
/ 11	primary	kappa=40.0
/ 23	primary	kappa=40.0
/ 37	primary	kappa=40.0
/ 11	positive	kappa=12.0
/ 23	positive	kappa=40.0
/ 37	positive	kappa=40.0

Parameter excerpt: first 12 of 72 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Cohorts differ across studies. A total out of 60 describes three trained models sharing 20 paired starts, not 60 independently trained models. Fixed/tuned settings, actual executions and aliases stay separate.

Evidence: outputs/overnight_research/pointmaze_critic_coverage_v1/COVERAGE_REPORT.txt; outputs/overnight_research/pointmaze_critic_coverage_v1/README.txt; table/source hashes are recorded in the snapshot and file catalog.

Local-H first feasibility rejection [ARCHIVED]

Task / success: Native MuJoCo PointMaze medium, repeated bounded 2D actions, actual successors and sparse task reward; 1000 native decisions unless explicitly modified.

Q: Learned IQL twin target-min Q and separate V, using actual (s,a,r,s_next,mask) tuples; exact update budgets differ by study.

BC: Learned BC flow conditioned on position x,y, with no explicit goal input. The full-data actor is frozen within a sampler comparison; task-specific Q/V provide goal preferences.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

No main comparison or policy test ran. A corrected 3,715-second resource bound exceeded the 2,400-second allowance. Original checker failures remain preserved.

Limit: No physical result or sampler ranking was obtained; only the resource-estimation and guard failures are preserved.

Evidence: outputs/overnight_research/pointmaze_local_H_v1/FEASIBILITY_REJECTION_REPORT.txt; outputs/overnight_research/pointmaze_local_H_v1/LOCAL_H_FEASIBILITY_REPORT.txt; outputs/overnight_research/pointmaze_local_H_v1/README.txt; table/source hashes are recorded in the snapshot and file catalog.

Local-H2 exposure follow-up [COMPLETE]

Task / success: Native MuJoCo PointMaze medium, repeated bounded 2D actions, actual successors and sparse task reward; 1000 native decisions unless explicitly modified.

Q: Learned IQL twin target-min Q and separate V, using actual (s,a,r,s_next,mask) tuples; exact update budgets differ by study.

BC: Learned BC flow conditioned on position x,y, with no explicit goal input. The full-data actor is frozen within a sampler comparison; task-specific Q/V provide goal preferences.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

There were 984 validation episodes and 660 actual tests: 33 physical configurations, 63 display rows and 30 aliases. Every guided FULL/HOLE condition achieved 60/60; BC achieved 0/60. All starts lay inside H, and the ceiling prevents ranking.

Task / seed	Method	Own frozen setting
/ 11	qgf	kappa=40.0
/ 11	qgf	kappa=40.0
/ 23	qgf	kappa=40.0
/ 23	qgf	kappa=40.0
/ 37	qgf	kappa=40.0
/ 37	qgf	kappa=40.0
/ 11	dflow_det_full_full__lambda0	kappa=40.0 eta=0.1
/ 11	dflow_det_full_full__lambda0	kappa=40.0 eta=0.1
/ 23	dflow_det_full_full__lambda0	kappa=40.0 eta=0.1
/ 23	dflow_det_full_full__lambda0	kappa=40.0 eta=0.1
/ 37	dflow_det_full_full__lambda0	kappa=40.0 eta=0.1
/ 37	dflow_det_full_full__lambda0	kappa=40.0 eta=0.1

Parameter excerpt: first 12 of 30 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Cohorts differ across studies. A total out of 60 describes three trained models sharing 20 paired starts, not 60 independently trained models. Fixed/tuned settings, actual executions and aliases stay separate.

Evidence: outputs/overnight_research/pointmaze_local_H_v2/LOCAL_H_V2_CONSUMER_REPORT.txt; outputs/overnight_research/pointmaze_local_H_v2/REPORT.txt; outputs/overnight_research/pointmaze_local_H_v2/owner_numeric_preflight/RESET_RECORD_SCOPE.txt; table/source hashes are recorded in the snapshot and file catalog.

First native endpoint-proxy geometry diagnosis [COMPLETE]

Task / success: See exact diagnostic unit in source; never add derived comparisons/chains to native episode totals.

Q: Saved learned target-Q arrays or frozen critic/analytic-score diagnostic; no new critic training.

BC: Same frozen actor/flow as parent study.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

Ninety-six conditions and 24,576 comparisons reused 768 distinct validation state/source/model pairs. There were no new physical steps. Directions were mixed/bimodal and proxy/full-Q values disagreed; this is not causal proof of rollout failure.

Task / seed	Method	Own frozen setting
/ 11	dflow_det_full_full	kappa=5.0 eta=0.1 iterations=256
/ 11	dflow_det_full_full	kappa=20.0 eta=0.03 iterations=64
/ 11	dflow_det_full_full	kappa=40.0 eta=0.01 iterations=256
/ 11	dflow_det_full_full	kappa=20.0 eta=0.03 iterations=64
/ 11	dflow_det_identity_full	kappa=12.0 eta=0.03 iterations=256
/ 11	dflow_det_identity_full	kappa=20.0 eta=0.03 iterations=64
/ 11	dflow_det_identity_full	kappa=40.0 eta=0.1 iterations=64
/ 11	dflow_det_identity_full	kappa=20.0 eta=0.03 iterations=64
/ 11	dflow_det_full_euler	kappa=40.0 eta=0.1 iterations=1
/ 11	dflow_det_full_euler	kappa=20.0 eta=0.03 iterations=64
/ 11	dflow_det_full_euler	kappa=0.2 eta=0.1 iterations=128
/ 11	dflow_det_full_euler	kappa=20.0 eta=0.03 iterations=64

Parameter excerpt: first 12 of 96 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Diagnostic target matching, replay and derived arrays are not independent trained models or ground-truth global Q error.

Evidence: outputs/overnight_research/endpoint_proxy_diagnostic/ENDPOINT_PROXY_REPORT.txt; outputs/overnight_research/endpoint_proxy_diagnostic/README.txt; table/source hashes are recorded in the snapshot and file catalog.

Saved twin-head disagreement diagnostic [COMPLETE]

Task / success: See exact diagnostic unit in source; never add derived comparisons/chains to native episode totals.

Q: Saved learned target-Q arrays or frozen critic/analytic-score diagnostic; no new critic training.

BC: Same frozen actor/flow as parent study.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

The analysis reused 4,080 saved tuples across six critics to form 24,480 derived records, with no new queries or episodes. Twin-head gaps did not consistently increase after support removal and are not calibrated Q uncertainty.

Limit: Diagnostic target matching, replay and derived arrays are not independent trained models or ground-truth global Q error.

Evidence: outputs/overnight_research/critic_disagreement_diagnostic/CRITIC_DISAGREEMENT_REPORT.txt; outputs/overnight_research/critic_disagreement_diagnostic/README.txt; table/source hashes are recorded in the snapshot and file catalog.

Paired descriptive native outcome analysis [COMPLETE]

Task / success: See exact diagnostic unit in source; never add derived comparisons/chains to native episode totals.

Q: Saved learned target-Q arrays or frozen critic/analytic-score diagnostic; no new critic training.

BC: Same frozen actor/flow as parent study.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

This reuses all 34 primary method rows. More successes can coexist with worse mean return. No new test-selected winner or additional episode is created.

Limit: Diagnostic target matching, replay and derived arrays are not independent trained models or ground-truth global Q error.

Evidence: outputs/overnight_research/paired_full_sampler_analysis/PAIRED_DESCRIPTIVE_REPORT.txt; table/source hashes are recorded in the snapshot and file catalog.

Residual-weighted IPG, CNF score and target controls [COMPLETE]

Task / success: See exact diagnostic unit in source; never add derived comparisons/chains to native episode totals.

Q: Saved learned target-Q arrays or frozen critic/analytic-score diagnostic; no new critic training.

BC: Same frozen actor/flow as parent study.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

Known-target weighting improved some sampling diagnostics, but selected six-task results were IPG 31/120, IPG-RW 30/120 and FK 65/120. Broad search stopped at 523 validation configurations. The score-provider control was mixed or negative.

Task / seed	Method	Own frozen setting
mode01 /	source record 1	flow_steps=32 particles=16
mode02 /	source record 2	flow_steps=32 particles=16
narrow01 /	source record 3	flow_steps=32 particles=16
curve01 /	source record 4	flow_steps=32 particles=16
missing01 /	source record 5	flow_steps=32 particles=16
hidden01 /	source record 6	flow_steps=32 particles=16
SIX_TASK_MEAN /	source record 7	flow_steps=32 particles=16
mode01 /	source record 8	flow_steps=128 particles=16
mode02 /	source record 9	flow_steps=128 particles=16
narrow01 /	source record 10	flow_steps=128 particles=16
curve01 /	source record 11	flow_steps=128 particles=16
missing01 /	source record 12	flow_steps=128 particles=16

Parameter excerpt: first 12 of 194 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Diagnostic target matching, replay and derived arrays are not independent trained models or ground-truth global Q error.

Evidence: outputs/overnight_research/sampler_audit/SAMPLER_AUDIT.md; outputs/overnight_research/sampler_audit/dflow_mh_agent/README.md; outputs/overnight_research/sampler_audit/pointmaze_geometry/README.txt; outputs/overnight_research/sampler_audit/score_provider_panel/NEGATIVE_FINDING.txt; table/source hashes are recorded in the snapshot and file catalog.

Fixed-target ULA/MH analytic panel [COMPLETE]

Task / success: See exact diagnostic unit in source; never add derived comparisons/chains to native episode totals.

Q: Fitted synthetic immediate-reward RBF critic; controlled omissions/false peaks, not IQL. True reward is separately known for validation/test.

BC: Analytic 2D behavior transport, not a learned neural BC policy.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

There were 432 endpoint conditions, with 512 independent chains within each condition. The iteration 16/64/256 values are correlated prefixes of those same chains. Some finite-step target errors improved with MH; these are toy chains, not native episodes.

Task / seed	Method	Own frozen setting
mode01 /	ULA	kappa=5.0 eta=0.01 iterations=16
mode01 /	ULA	kappa=5.0 eta=0.01 iterations=64
mode01 /	ULA	kappa=5.0 eta=0.01 iterations=256
mode01 /	MH_corrected	kappa=5.0 eta=0.01 iterations=16
mode01 /	MH_corrected	kappa=5.0 eta=0.01 iterations=64
mode01 /	MH_corrected	kappa=5.0 eta=0.01 iterations=256
mode01 /	ULA	kappa=5.0 eta=0.03 iterations=16
mode01 /	ULA	kappa=5.0 eta=0.03 iterations=64
mode01 /	ULA	kappa=5.0 eta=0.03 iterations=256
mode01 /	MH_corrected	kappa=5.0 eta=0.03 iterations=16
mode01 /	MH_corrected	kappa=5.0 eta=0.03 iterations=64
mode01 /	MH_corrected	kappa=5.0 eta=0.03 iterations=256

Parameter excerpt: first 12 of 432 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Diagnostic target matching, replay and derived arrays are not independent trained models or ground-truth global Q error.

Evidence: ; table/source hashes are recorded in the snapshot and file catalog.

Twenty-task ULA/MH independent sweep [COMPLETE]

Task / success: See exact diagnostic unit in source; never add derived comparisons/chains to native episode totals.

Q: Fitted synthetic immediate-reward RBF critic; controlled omissions/false peaks, not IQL. True reward is separately known for validation/test.

BC: Analytic 2D behavior transport, not a learned neural BC policy.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

The MH/ULA sweep produced 96,000 validation rows and 3,200 new MH/ULA test endpoints. Full/full MH achieved 181/400 and ULA 151/400; fresh QGF achieved 154/400. Lambda=T=1. Separate BC/QGF reference points in the method summary are not part of the 3,200-endpoint count.

Task / seed	Method	Own frozen setting
mode01 /	BC_exact	See exact source row / original table
mode01 /	QGF_0_prior_control	kappa=0.0
mode01 /	QGF_0.02	kappa=0.02
mode01 /	QGF_0.04	kappa=0.04
mode01 /	QGF_tuned	kappa=3.0
mode02 /	BC_exact	See exact source row / original table
mode02 /	QGF_0_prior_control	kappa=0.0
mode02 /	QGF_0.02	kappa=0.02
mode02 /	QGF_0.04	kappa=0.04
mode02 /	QGF_tuned	kappa=3.0
mode03 /	BC_exact	See exact source row / original table
mode03 /	QGF_0_prior_control	kappa=0.0

Parameter excerpt: first 12 of 260 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Diagnostic target matching, replay and derived arrays are not independent trained models or ground-truth global Q error.

Evidence: outputs/overnight_research/dflow_mh_sweep/DFLOW_MH_REPORT.txt; table/source hashes are recorded in the snapshot and file catalog.

Privileged robot actor repair and pose follow-up [COMPLETE]

Task / success: Native cube-triple robot: yellow cube to its goal while other cubes remain within their goals; all three centers within 4 cm, up to 1000 decisions. Native reward is number of correctly placed cubes minus 3; success can occur while a cube is still held.

Q: Q, target-Q and V stayed frozen and were not queried. This privileged actor/pose repair evaluated BC actions without critic guidance.

BC: Frozen or repaired privileged-demo BC actor, as labeled in the original stage. No critic guidance is used in the stated actor control.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

The first selected actor repair achieved 2/20 versus original BC 0/20, but mean return worsened to about -1,232 versus -1,000. A separate pose-diverse control gave old 3/20 versus new 0/20. These comparisons used no critic guidance.

Task / seed	Method	Own frozen setting
/	old12_demo_pool	kappa=0.0 flow_steps=8 particles=1 noise_strength=0.0 ridge=0.1
/	new44_demo_pool	kappa=0.0 flow_steps=8 particles=1 noise_strength=0.0 ridge=0.1
/	original	kappa=0.0 flow_steps=8 particles=1 noise_strength=0.0 ridge=0.1
/	oracle90	kappa=0.0 flow_steps=8 particles=1 noise_strength=0.0 ridge=0.1

Parameter excerpt: first 4 of 4 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Original logged data and the privileged 44-demo / 1,916-tuple augmentation are labeled separately. Capacity, replanning and temporal-output changes are not fully isolated.

Evidence: outputs/overnight_research/robot_repair/ROBOT_REPAIR_REPORT.txt; outputs/overnight_research/robot_repair/pose_study/POSE_STUDY_REPORT.txt; table/source hashes are recorded in the snapshot and file catalog.

Matched H1/H5 robot action-chunk study [COMPLETE]

Task / success: Native cube-triple robot: yellow cube to its goal while other cubes remain within their goals; all three centers within 4 cm, up to 1000 decisions. Native reward is number of correctly placed cubes minus 3; success can occur while a cube is still held.

Q: No Q, target-Q or V updates or guidance occurred. Unused networks remained fresh/canonical; this is privileged-demo actor-only BC, not a learned-IQL sampler comparison.

BC: Frozen or repaired privileged-demo BC actor, as labeled in the original stage. No critic guidance is used in the stated actor control.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

Across three trained seeds, H1 success counts were 1, 3 and 0 per 20; H5 counts were 6, 3 and 2 per 20. The totals were 4/60 and 11/60. H5 had two collateral-damage episodes. This was an actor-only BC comparison, not a guided-Q comparison.

Limit: Original logged data and the privileged 44-demo / 1,916-tuple augmentation are labeled separately. Capacity, replanning and temporal-output changes are not fully isolated.

Evidence: outputs/overnight_research/robot_horizon_control/HORIZON_STUDY_REPORT.txt; outputs/overnight_research/robot_horizon_control/aborted_step_dtype_prephysics/README.txt; table/source hashes are recorded in the snapshot and file catalog.

Proper 25D H5 IQL and terminal-invariance repair [COMPLETE]

Task / success: Native cube-triple robot: yellow cube to its goal while other cubes remain within their goals; all three centers within 4 cm, up to 1000 decisions. Native reward is number of correctly placed cubes minus 3; success can occur while a cube is still held.

Q: Learned IQL target-min Q and V on actual robot transitions; horizon1 has 5D actions, horizon5 critic has 25D discounted action prefixes.

BC: Learned robot BC flow; privileged successful-demo augmentation is explicitly separated from original logged data.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

Known-goal uniform-action RMSE improved from 86.35 to 0.707, yet all six actual physical settings remained at 2/20, for 120 actual tests. Calibration repair did not restore grasp competence.

Task / seed	Method	Own frozen setting
/	parent50k	kappa=0.0
/	parent50k	kappa=0.0
/	parent50k	kappa=0.0
/	parent50k	kappa=0.002
/	parent50k	kappa=0.002
/	parent50k	kappa=0.002
/	parent50k	kappa=0.01
/	parent50k	kappa=0.01
/	parent50k	kappa=0.01
/	parent50k	kappa=0.04
/	parent50k	kappa=0.04
/	parent50k	kappa=0.04

Parameter excerpt: first 12 of 78 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Original logged data and the privileged 44-demo / 1,916-tuple augmentation are labeled separately. Capacity, replanning and temporal-output changes are not fully isolated.

Evidence: outputs/overnight_research/robot_chunk_critic/CHUNK_CRITIC_TRAINING_REPORT.txt; outputs/overnight_research/robot_chunk_critic/critic_actor_alignment/ALIGNMENT_DIAGNOSTIC_REPORT.txt; outputs/overnight_research/robot_chunk_critic/physical_critic_comparison/PHYSICAL_CRITIC_COMPARISON_REPORT.txt; outputs/overnight_research/robot_chunk_critic/terminal_invariance_repair/TERMINAL_INVARIANCE_REPORT.txt; table/source hashes are recorded in the snapshot and file catalog.

Known narrow/broad false-Q versus genuine optima [COMPLETE]

Task / success: One-dimensional analytic action a=tanh(z), z drawn from N(0,1). Success means |a-good_center| <= success_radius: a target interval, not a 2D circle. There is no next state or learned IQL critic.

Q: Analytic known true and artificial score maps, not learned IQL.

BC: Known one-dimensional source/transport, not a learned neural BC actor.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

There were 1,440 distinct heldout action points. Selected regular, stochastic and MH families achieved 107/240, 76/240 and 126/240; their Q-averaged versions achieved 119/240, 81/240 and 116/240.

Task / seed	Method	Own frozen setting
narrow_false_peak_0 /	source record 1	kappa=20.0 eta=0.005 iterations=256 sigma=0.0 K=1
narrow_false_peak_0 /	source record 2	kappa=20.0 eta=0.005 iterations=256 sigma=0.0 K=1
narrow_false_peak_0 /	source record 3	kappa=20.0 eta=0.005 iterations=256 sigma=0.0 K=1
narrow_false_peak_0 /	source record 4	kappa=20.0 eta=0.005 iterations=256 sigma=0.0 K=1
narrow_false_peak_0 /	source record 5	kappa=5.0 eta=0.1 iterations=64 sigma=0.0 K=1
narrow_false_peak_0 /	source record 6	kappa=5.0 eta=0.1 iterations=64 sigma=0.0 K=1
narrow_false_peak_0 /	source record 7	kappa=5.0 eta=0.03 iterations=256 sigma=0.15 K=16
narrow_false_peak_0 /	source record 8	kappa=5.0 eta=0.03 iterations=256 sigma=0.15 K=16
narrow_false_peak_0 /	source record 9	kappa=20.0 eta=0.005 iterations=256 sigma=0.15 K=16
narrow_false_peak_0 /	source record 10	kappa=20.0 eta=0.005 iterations=256 sigma=0.15 K=16
narrow_false_peak_0 /	source record 11	kappa=5.0 eta=0.03 iterations=256 sigma=0.05 K=4
narrow_false_peak_0 /	source record 12	kappa=5.0 eta=0.03 iterations=256 sigma=0.05 K=4

Parameter excerpt: first 12 of 144 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Each family selects its own hyperparameters. Smoothing narrow false peaks need not help broad critic error or genuine narrow optima.

Evidence: outputs/overnight_research_20261105/controlled_scores/REPORT.txt; table/source hashes are recorded in the snapshot and file catalog.

Matched-target quadrature and finite-chain diagnosis [COMPLETE]

Task / success: Two hundred forty independent source/case bundles form 960 paired one-dimensional chains. Five prefixes reuse the same chains and are correlated. These are not native episodes.

Q: Known one-dimensional raw and Q-averaged analytic targets, with stationary quadrature and matched frozen positive MH settings.

BC: Known one-dimensional source/transport, not a learned neural BC actor.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

Finite-prefix results differed substantially from target mass, and broad false-region targets remained misleading. Quadrature convergence error was 1.07e-5.

Task / seed	Method	Own frozen setting
narrow_false_peak_0 /	ULA	kappa=5.0 eta=0.1 iterations=1 sigma=0.0 K=1
narrow_false_peak_0 /	ULA	kappa=5.0 eta=0.1 iterations=4 sigma=0.0 K=1
narrow_false_peak_0 /	ULA	kappa=5.0 eta=0.1 iterations=16 sigma=0.0 K=1
narrow_false_peak_0 /	ULA	kappa=5.0 eta=0.1 iterations=64 sigma=0.0 K=1
narrow_false_peak_0 /	ULA	kappa=5.0 eta=0.1 iterations=256 sigma=0.0 K=1
narrow_false_peak_0 /	MH	kappa=5.0 eta=0.1 iterations=1 sigma=0.0 K=1
narrow_false_peak_0 /	MH	kappa=5.0 eta=0.1 iterations=4 sigma=0.0 K=1
narrow_false_peak_0 /	MH	kappa=5.0 eta=0.1 iterations=16 sigma=0.0 K=1
narrow_false_peak_0 /	MH	kappa=5.0 eta=0.1 iterations=64 sigma=0.0 K=1
narrow_false_peak_0 /	MH	kappa=5.0 eta=0.1 iterations=256 sigma=0.0 K=1
narrow_false_peak_0 /	ULA	kappa=5.0 eta=0.03 iterations=1 sigma=0.05 K=4
narrow_false_peak_0 /	ULA	kappa=5.0 eta=0.03 iterations=4 sigma=0.05 K=4

Parameter excerpt: first 12 of 240 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: Sampling the chosen target more faithfully cannot make a wrong Q target correct. The five prefixes are correlated, not 4,800 independent chains.

Evidence: outputs/overnight_research_20261105/matched_target_diagnostic/v2/REPORT.txt; table/source hashes are recorded in the snapshot and file catalog.

Second-night all-12-family FULL/HOLE coverage comparison [COMPLETE]

Task / success: Native PointMaze dynamics with a modified TRAIN-position reset mapping and a validation-selected 128-decision cap. This is distinct from the standard 1,000-decision task.

Q: FULL/HOLE frozen IQL Q/V critics, three trainedseeds; critic-onlycoverageomission, actualtuplespreserved.

BC: Learned BC flow conditioned on position x,y, with no explicit goal input. The full-data actor is frozen within a sampler comparison; task-specific Q/V provide goal preferences.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

All 174 choices froze before testing. The ledger records 4,560 actual test executions, including 100 numeric duplicates (4,460 after that deduplication), and 11,360 validation executions including 48 numeric duplicates. QGF FULL successes were 20, 13 and 20 per 20; HOLE successes were 20, 19 and 20 per 20.

Task / seed	Method	Own frozen setting
/ 11	bc	Canonical BC; Q/source-loop fields unused
/ 11	bc_particles_N8_T8	particles=8 flow_steps=8
/ 11	bc_particles_N8_T32	particles=8 flow_steps=32
/ 11	bc_particles_N16_T8	particles=16 flow_steps=8
/ 11	bc_particles_N16_T32	particles=16 flow_steps=32
/ 11	qgf_fixed_0.02	kappa=0.02
/ 11	qgf_fixed_0.04	kappa=0.04
/ 11	greedy_q_grid	81-action Q argmax; kappa/eta/lambda/I unused
/ 11	qgf_fixed_40	kappa=40.0
/ 11	qgf	kappa=40
/ 11	dflow_det_full_full__lambda0	kappa=80 eta=0.1 lambda=0.0 iterations=256
/ 11	dflow_det_full_full__lambda1	kappa=80 eta=0.03 lambda=1.0 iterations=256

Parameter excerpt: first 12 of 230 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: TRAIN-origin reset anchors are not entirely new learning-state support. BC has full coverage; starts are paired and compute is unequal. No universal winner follows.

Evidence: outputs/overnight_research_20261105/coverage_stress/comparison/independent_replay/README.txt; outputs/overnight_research_20261105/coverage_stress/comparison/independent_replay/prepared_v1_archive/README.txt; outputs/overnight_research_20261105/coverage_stress/comparison/independent_replay/prepared_v2_archive/README.txt; outputs/overnight_research_20261105/coverage_stress/comparison/independent_replay/prepared_v3_initial_watchdog_archive/README.txt; table/source hashes are recorded in the snapshot and file catalog.

Second-night actor Jacobian and clipping diagnostic [COMPLETE]

Task / success: Forty-eight Gaussian draws, 96 paired scales, 1,536 derived Jacobian rows and 192 FD checks; no environment episodes.

Q: Actor-only endpoint/Jacobian geometry; no Q queries or true-Q accuracy labels.

BC: Learned conditional BC flow. Goal-specific critic changes; full-data actor is fixed within each sampler comparison.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

At unit source scale, Euler sensitivity was attenuated but nonzero. Tail-scale sources clipped more often. Applicable original FD checks passed. The report keeps raw and clipped Jacobians separate.

Limit: Paired source scales and shared actors are not additional trained seeds. Later derivatives are with respect to current x_t, not the initial source. Last-step equality is not an identity Jacobian. No critic-direction or rollout-causality proof follows.

Evidence: outputs/overnight_research_20261105/endpoint_jacobian_diagnostic/REPORT.txt; outputs/overnight_research_20261105/endpoint_jacobian_diagnostic/REPORT_INSERT.txt; table/source hashes are recorded in the snapshot and file catalog.

Current standard task1 continuation [COMPLETE]

Task / success: Native MuJoCo PointMaze medium, repeated bounded 2D actions, actual successors and sparse task reward; 1000 native decisions unless explicitly modified. Task1near(0,0)->(20,20).

Q: Learned IQL twin target-min Q and separate V, using actual (s,a,r,s_next,mask) tuples; exact update budgets differ by study. Fixed50k critics.

BC: Learned BC flow conditioned on position x,y, with no explicit goal input. The full-data actor is frozen within a sampler comparison; task-specific Q/V provide goal preferences.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

Verified saved JSON fact	Value
Actual active episodes	1620
Archived Euler episodes	500
Archived CV/FK episodes	80
All historical episodes	2200
Additional B20 episodes	0

All three standard task-1 models completed the original 20-start cohort. The independently checked execution ledger records 2,200 actual historical test episodes: 1,620 active full-flow/QGF/IPG/control episodes, 500 archived Euler-endpoint episodes and 80 archived IPG-CV/FK episodes. The active 27 primary requests per model have no aliases or positive extras. Raw QGF scored 13/20, 20/20 and 20/20; regular full-J/full-flow lambda=1 scored 16/20, 20/20 and 20/20.

Task / seed	Method	Own frozen setting
1 / 37	bc	Canonical BC; Q/source-loop fields unused
1 / 37	bc_particles_N8_T8	particles=8 flow_steps=8
1 / 37	bc_particles_N8_T32	particles=8 flow_steps=32
1 / 37	bc_particles_N16_T8	particles=16 flow_steps=8
1 / 37	bc_particles_N16_T32	particles=16 flow_steps=32
1 / 37	qgf_fixed_0.02	kappa=0.02
1 / 37	qgf_fixed_0.04	kappa=0.04
1 / 37	greedy_q_grid	81-action Q argmax; kappa/eta/lambda/I unused
1 / 37	qgf	kappa=40.0
1 / 37	dflow_det_full_full__lambda0	kappa=5.0 eta=0.1 lambda=0.0 iterations=256
1 / 37	dflow_det_full_full__lambda1	kappa=80.0 eta=0.03 lambda=1.0 iterations=256
1 / 37	dflow_det_full_euler__lambda0	Canonical BC; Q/source-loop fields unused

Parameter excerpt: first 12 of 252 source-order selected/evaluated result settings. Every row is in ALL_EXPERIMENTS_SELECTED_PARAMETERS.csv; parameter_view_kind distinguishes explicit selections from fixed/evaluated controls. ALL_EXPERIMENTS_CONFIG_FIELDS.csv retains every source JSON config field, and the ZIP includes the verbatim files. Overlapping views are not extra trials.

Limit: The original 20-start cohort is preserved. The target of 40 starts has not been executed in this evidence set. All 87 legacy choices froze before task-1 tests. Euler/CV/FK outcomes and their costs remain archived.

Evidence: ; table/source hashes are recorded in the snapshot and file catalog.

Current standard task5 completed seed11 validation [PARTIAL]

Task / success: Native MuJoCo PointMaze medium, repeated bounded 2D actions, actual successors and sparse task reward; 1000 native decisions unless explicitly modified. Task5near(20,4)->(0,0).

Q: Learned IQL twin target-min Q and separate V, using actual (s,a,r,s_next,mask) tuples; exact update budgets differ by study. Goal-specific50k.

BC: Learned BC flow conditioned on position x,y, with no explicit goal input. The full-data actor is frozen within a sampler comparison; task-specific Q/V provide goal preferences.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

At the raw inventory snapshot, seed 11 had completed 224 B8 batches (1,792 cumulative validation episodes), 27 legacy choices and zero heldout tests. The other models were unfinished. No task-5 heldout sampler ranking is available in this fixed evidence set.

Limit: At the raw inventory snapshot, seed 23 was running full-flow validation and seed 37 was pending the retained 19 conditions. The raw QGF seed-23 validation baseline of 3/8 must remain visible.

Evidence: ; table/source hashes are recorded in the snapshot and file catalog.

Current Euler/lambda wiring and geometry results [COMPLETE]

Task / success: Sixfrozen task/modelcells;192pairedscalestrata,192FDrows,192one-stepcomparisons;0physics.

Q: Samelearned IQLtarget-minQ; diagnosticdoesnotmeasuretrueQerror.

BC: Learned conditional BC flow. Goal-specific critic changes; full-data actor is fixed within each sampler comparison.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

All 192 one-step action-parity and 192 endpoint-FD comparisons passed; 31 Q-score-FD misses remain. Euler and full-F8 source gradients can point differently. The source-prior sign was negative on the declared probes.

Limit: One-step action parity does not prove every outer iteration and may mask source errors under saturation. The user excluded future D-Flow Euler-proxy work; original QGF per-slice semantics remain unchanged.

Evidence:
outputs/overnight_research_20261106_continuation/review/Euler_diagnostic_results/20261106T185547_025737Z/REPORT.txt; outputs/overnight_research_20261106_continuation/review/Euler_diagnostic_results/20261106T185547_025737Z/REPORT_INSERT.txt; table/source hashes are recorded in the snapshot and file catalog.

Precision-only discriminator [COMPLETE]

Task / success: Exactsame192Euler/F8and16addonactionFDnodes/banks, unchangedh/tolerances, storedf32weights;64-bitinferenceonly.

Q: SamefixedIQL;0trainingorparametercasts.

BC: Learned conditional BC flow. Goal-specific critic changes; full-data actor is fixed within each sampler comparison.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

All 208 fixed-node float64 FD comparisons passed. Maximum source AD32/AD64 difference was 2.681e-7; the addon difference was 1.052e-7. Original 31+1 failed float32 checks remain preserved.

Limit: The observed reduction supports roundoff sensitivity at these fixed points, not critic accuracy or correctness of every code path. The addon direct B2 x K reference is not production fixed-B20 graph identity.

Evidence:
outputs/overnight_research_20261106_continuation/review/precision_FD_results/20261106T191212_571252Z/REPORT.txt;
outputs/overnight_research_20261106_continuation/review/precision_FD_results/20261106T191212_571252Z/REPORT_INSERT.txt;
table/source hashes are recorded in the snapshot and file catalog.

New Q-averaged stochastic identity/full-flow prototype [PARTIAL]

Task / success: Actual full F8 forward endpoint with identity backward guidance. A fixed whole-call K=4/16 antithetic panel perturbs final actions. Engineering checks alone do not establish task success.

Q: The same frozen task-specific IQL target-min critic is averaged over nearby clipped final actions. The V2 result is numerical engineering only, with zero native episodes.

BC: Learned conditional BC flow. Goal-specific critic changes; full-data actor is fixed within each sampler comparison.

Split and selection: Study-specific source protocol is authoritative. Training, validation, heldout cohorts and diagnostic reuses remain separate; selected settings are not globally optimal.

Training and sampler budgets: Exact study-specific training configs, update budgets, candidate grids and frozen selections are included verbatim in the listed source/config evidence. Selected sampler settings are distinct from training updates and physical decisions; no common budget is inferred across robots, custom navigation and native PointMaze.

Verified saved JSON fact	Value
Raw checks	105
Raw failures retained	1
Eligible FD64 comparisons	16
Production AD/reference comparisons	16
Qualified failures	0
Native/test episodes	0

The original engineering run retained 104/105 raw checks passing and one float32 Qbar-FD failure. The separate precision discriminator passed all 16 saved-bank addon reference checks. The reviewed V2 task-1/seed-11 gate then passed 89 unchanged non-FD checks, 16 eligible FD64 checks and 16 production-AD/reference comparisons, while retaining raw FAIL 1/105. It used 15.89 seconds and no native episodes; profile and physical evaluation were not run in that audited result.

Limit: V1 raw FAIL and all source/arrays remain preserved. V2 qualifies a separate engineering acceptance rule with byte-identical production kernel; it is the same D-Flow family, not a physical-result comparison.

Evidence: ; table/source hashes are recorded in the snapshot and file catalog.

Pending work, failures and evidence limits

Fresh B20 cohorts for the 40-start target are pending in this fixed evidence set. Original A20 scores remain separate.

Task 5 had only partial completed-model validation at the raw inventory timestamp. This report does not infer later completion or a heldout ranking.

The new identity-Q-averaged stochastic full-flow family has qualified numerical engineering evidence, but no native profile/validation/test result in this fixed evidence set.

Future D-Flow uses full-flow endpoints only. Historical Euler-endpoint and IPG-CV/FK results and their compute remain archived. Original QGF per-slice Euler guidance stays unchanged.

This is a fixed as-of historical report. No global trial sum is computed across overlapping summaries, reused cohorts, replays, aliases, particle populations and diagnostics.

The target is 40 episodes per method and trained model, using a preserved 20 plus 20 fresh paired starts with exactly frozen settings. This report prints original successes/20 until the second cohort actually completes.

The current full-flow-only D-Flow scope is nine active families / 16 lambda conditions, including the separately exploratory identity-Q-averaged stochastic family. The main retained core is 19 guided conditions plus eight controls; the new family has two separate lambda conditions.

The new evidence ZIP is a compact report/table/source companion. It references the existing reproducible archives and does not replace their datasets, checkpoints or runtimes. The archive recheck confirms full-byte SHA matches and stable metadata; it is not a new CRC/extraction verification.

The original float32 Euler/lambda and exploratory addon engineering failures remain preserved. The separately identified fixed-node float64 discriminator and V2 qualification are numerical evidence, not proof of critic accuracy or task causality. An accepted qualified gate does not relabel the original raw checks as passing and is not another D-Flow family or a physical success.

Three trained seeds, paired starts, heterogeneous inference budgets, privileged successful-demo augmentation in some repairs, finite validation searches and critic extrapolation restrict conclusions. Completed zeros and unsuccessful repairs remain evidence. Synthetic points, modified navigation and MuJoCo manipulation are reported separately.

Exact fixed input set: 969 files / 9,443,441 bytes. Mutable progress/metrics and live partial pickle contents are excluded. Narrative findings are supplied by the root catalog and tied to its listed evidence; JSON facts are checked against their precise source values. Source hashes and metadata establish provenance; they do not rerun inference or independently prove a causal explanation.

Explicit skipped and pending inputs

live_work [MUTABLE_EXCLUDED]: Live progress, growing candidate metrics, uncompleted test files and partial pickle contents are excluded; only the explicitly hashed fixed evidence set is collected.

SYNTHETIC OFFLINE-CRITIC EXPERIMENT | 4 NOVEMBER 2026

D-Flow and QGF: completed comparison

What is the task? These are artificial "choose a point" puzzles. Each action is a point (x,y). A hidden scoring function rewards some locations more than others. The method succeeds when its chosen point is close enough to the specified good target. There is no robot or maze in the counts in this report.

20 task configurations, 20 fresh test starts per condition, 8 D-Flow variants, and both lambda settings. Each fixed lambda condition gets its own kappa, eta and iteration selection on 40 separate validation starts. All reported counts below are held out.

This is a two-dimensional, one-step bandit diagnostic with an analytical reference flow and a fitted critic. It is not a reproduction of the QGF paper or a multi-step RL benchmark.

D-Flow variant	lambda = 0	lambda = 1
Regular / full gradient / full endpoint	201/400	209/400
Regular / identity gradient / full endpoint	201/400	201/400
Regular / full gradient / Euler endpoint	126/400	171/400
Regular / identity gradient / Euler endpoint	128/400	155/400
Stochastic / full gradient / full endpoint	105/400	172/400
Stochastic / identity gradient / full endpoint	164/400	191/400
Stochastic / full gradient / Euler endpoint	93/400	149/400
Stochastic / identity gradient / Euler endpoint	113/400	146/400

Reference	Held-out success	Mean reward
Behavior reference (BC)	56/400	0.256
QGF, fixed kappa = 0.02	60/400	0.264
QGF, fixed kappa = 0.04	62/400	0.274
QGF, independently tuned kappa	166/400	0.570

Observed pattern: lambda = 1 improved the pooled success of seven D-Flow variants and tied one. The strongest observed stochastic condition reached 191/400; regular full-gradient/full-endpoint reached 209/400. Fixed small-kappa QGF is much weaker here than tuned QGF, so the tuned reference is essential.

WHAT WAS ACTUALLY RUN

Tasks, data and tuning protocol

Think of a map with good and bad places to put a dot. Q predicts the score for each possible dot, using examples of dots and their observed scores. D-Flow and QGF use that prediction to move their proposed dot. The five families below change the good locations, the examples available, or the errors in Q.

Family: four variants each	Task and coverage design
Mode selection	Two reward modes; change the better side and reference-flow offset.
Narrow and sparse targets	Smaller reward peaks and tighter success regions, including shifted vertical targets.
Curved action distributions	Bending and wavy transports couple the two action coordinates.
Missing coverage	Remove a source tail or a ball near a useful target from the offline observations. The reference prior can still generate those actions.
Controlled hidden OOD error	Remove an action-space ball, then inject a smooth false critic peak strictly inside it. The injected error is exactly zero at every observed training action.

Offline data and critic. Each task uses 650-850 fixed action/reward observations with small Gaussian reward noise. A smooth RBF-plus-affine critic is fitted by ridge regression. Rewards are known mixtures of peaks with a weak quadratic tail penalty. Q equals expected immediate reward in this terminal bandit; no Bellman bootstrapping is used.

Reference flow. A known invertible two-dimensional triangular transport creates multimodal and curved action distributions. The full transport and its Jacobian are analytical. The continuous path is H(t,z) = [1-h(t)]z + h(t)F(z), with h(t) = (t+t^2)/2. This makes the source Euler endpoint an approximation rather than exactly F(z).

Setting	Values / definition
Kappa: Q correction strength	0, 0.02, 0.04, 0.06, 0.08, 0.1, 0.2, 0.3, 0.4, 0.6, 0.8, 1, 2, 3, 5
Eta: D-Flow optimizer step size	0.005, 0.01, 0.03, 0.1
D-Flow iterations	1, 2, 4, 8, 16, 32, 64, 128, 256
Lambda and stochastic noise	Lambda = 0 and 1; stochastic temperature T = 1. Regular variants omit the noise term.
QGF	32 integration steps; fixed kappa = 0.02 and 0.04; an additional reference tunes kappa on the full grid.

Selection: independently maximize validation success count, then mean validation reward; ties prefer fewer iterations and smaller coefficients. The 40 validation starts use true simulator reward. The 20 fresh test starts never select settings. All methods share the same initial seeds within a split.

LAMBDA CHOSEN ON VALIDATION, NEVER ON TEST RESULTS

Every task: successes out of 20

D = regular; S = stochastic. First letter: F = full gradient, I = identity gradient. Second letter: F = full endpoint, E = Euler endpoint. D-Flow cells show successes/20 [chosen lambda]. All selected kappa, eta and iterations are supplied in the CSV tables.

Task	BC	D FF	D IF	D FE	D IE	S FF	S IF	S FE	S IE	Q .02	Q .04	Q tuned
Right mode	5/20	12/20 [0]	12/20 [0]	12/20 [1]	12/20 [1]	12/20 [1]	11/20 [1]	14/20 [1]	13/20 [1]	5/20	5/20	11/20
Left mode	5/20	13/20 [0]	13/20 [0]	12/20 [1]	12/20 [1]	13/20 [1]	13/20 [1]	14/20 [1]	11/20 [1]	5/20	5/20	13/20
Offset right mode	4/20	9/20 [0]	9/20 [0]	8/20 [1]	8/20 [1]	12/20 [1]	12/20 [1]	10/20 [1]	13/20 [1]	4/20	4/20	8/20
Offset left mode	2/20	11/20 [0]	11/20 [0]	10/20 [1]	8/20 [1]	16/20 [1]	12/20 [1]	10/20 [1]	14/20 [1]	2/20	2/20	10/20
Narrow right	1/20	9/20 [1]	8/20 [0]	4/20 [1]	4/20 [1]	2/20 [1]	2/20 [1]	4/20 [1]	3/20 [1]	1/20	1/20	4/20
Narrow left	2/20	14/20 [1]	14/20 [1]	9/20 [1]	9/20 [1]	4/20 [1]	7/20 [1]	4/20 [1]	3/20 [1]	3/20	3/20	9/20
Sparse upper right	0/20	10/20 [1]	8/20 [1]	5/20 [1]	5/20 [1]	4/20 [1]	3/20 [0]	2/20 [1]	1/20 [1]	0/20	0/20	4/20
Sparse lower left	0/20	7/20 [1]	7/20 [1]	4/20 [1]	4/20 [0]	4/20 [1]	4/20 [1]	3/20 [1]	3/20 [1]	0/20	0/20	4/20
Rising curve	3/20	11/20 [1]	11/20 [0]	9/20 [1]	8/20 [1]	13/20 [1]	8/20 [1]	6/20 [1]	5/20 [1]	4/20	4/20	8/20
Falling curve	9/20	12/20 [1]	12/20 [0]	14/20 [1]	10/20 [1]	12/20 [1]	9/20 [1]	3/20 [1]	9/20 [1]	9/20	10/20	11/20
Wavy rising curve	2/20	12/20 [1]	9/20 [0]	5/20 [1]	5/20 [0]	12/20 [1]	9/20 [0]	10/20 [1]	8/20 [1]	2/20	2/20	5/20
Wavy falling curve	3/20	8/20 [0]	8/20 [0]	7/20 [1]	7/20 [1]	11/20 [1]	7/20 [1]	9/20 [1]	8/20 [1]	3/20	4/20	7/20
Unobserved right tail	0/20	8/20 [0]	8/20 [0]	9/20 [1]	6/20 [0]	7/20 [1]	10/20 [1]	7/20 [1]	5/20 [1]	1/20	1/20	4/20
Unobserved left tail	2/20	8/20 [0]	8/20 [0]	6/20 [0]	5/20 [0]	5/20 [1]	16/20 [1]	6/20 [1]	10/20 [1]	2/20	2/20	7/20
Missing upper target	0/20	11/20 [1]	11/20 [1]	9/20 [1]	9/20 [1]	5/20 [1]	3/20 [1]	2/20 [1]	0/20 [0]	0/20	0/20	8/20
Missing lower target	2/20	12/20 [1]	12/20 [1]	9/20 [1]	6/20 [1]	1/20 [1]	7/20 [1]	1/20 [0]	3/20 [1]	2/20	2/20	12/20
False upper peak	7/20	15/20 [0]	15/20 [0]	16/20 [1]	16/20 [1]	14/20 [1]	15/20 [1]	14/20 [1]	14/20 [1]	7/20	7/20	15/20
False lower peak	2/20	6/20 [1]	6/20 [1]	5/20 [1]	5/20 [1]	15/20 [1]	15/20 [1]	7/20 [1]	6/20 [1]	3/20	3/20	5/20
False right peak	4/20	10/20 [1]	10/20 [1]	8/20 [1]	8/20 [1]	6/20 [1]	14/20 [1]	9/20 [1]	10/20 [1]	4/20	4/20	9/20
False left peak	3/20	12/20 [1]	12/20 [0]	12/20 [1]	12/20 [1]	4/20 [1]	13/20 [1]	11/20 [1]	7/20 [1]	3/20	3/20	12/20

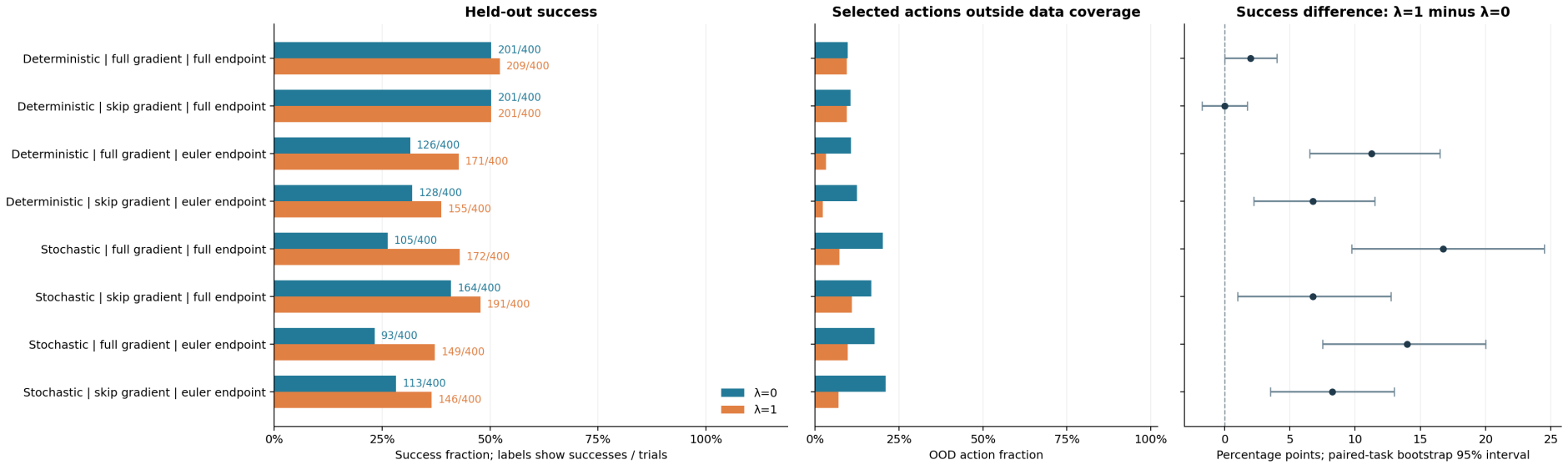
No tuned-QGF task has zero successes. Fixed small-kappa QGF gets 0/20 on three tasks, but tuning recovers 4/20, 4/20 and 8/20. These observations do not establish that QGF cannot solve the tasks.

OPTIONAL SHRINKING OF THE SOURCE NOISE INPUT

What lambda changes

D-Flow edits a random seed z and then generates the action F(z). Lambda multiplies an extra pull toward zero. At eta = 0.01, lambda = 1 replaces the old z contribution with 0.99z before adding the Q push and any noise. Lambda = 0 omits that shrinking. It does not measure or repair Q-function errors.

λ=1 adds a pull on the noise input toward zero



λ controls the optional pull toward the Gaussian source center; it does not correct Q-function errors. Each λ condition independently tunes κ, η and iterations on validation starts. This figure compares tuned conditions, rather than isolating a fixed-parameter causal effect of λ. Synthetic analytic-flow tasks; intervals exclude critic-training uncertainty.

Reading the comparison: the two lambda conditions independently retune kappa, eta and iterations. The figure therefore compares tuned conditions, rather than a fixed-parameter causal effect of lambda. OOD means low reference density or explicitly omitted coverage; a good missing-coverage target can itself be OOD.

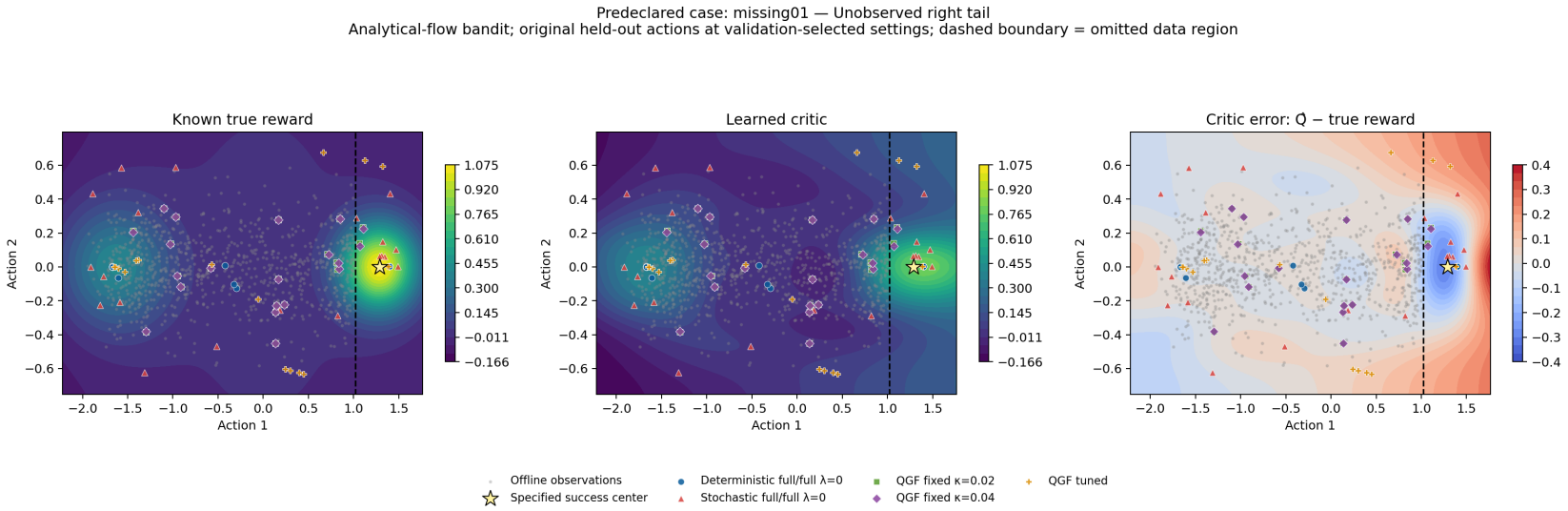
Exact update: $g(z) = J_E(z)^T$ grad Q(E(z)) for the full gradient, or grad Q(E(z)) for the identity gradient. Regular: z_next = z + eta [kappa g(z) - lambda z]. Stochastic: add sqrt(2 eta T) times a standard Gaussian draw. E is either F(z) or z+v(z,0). Final D-Flow actions always use F(z_final).

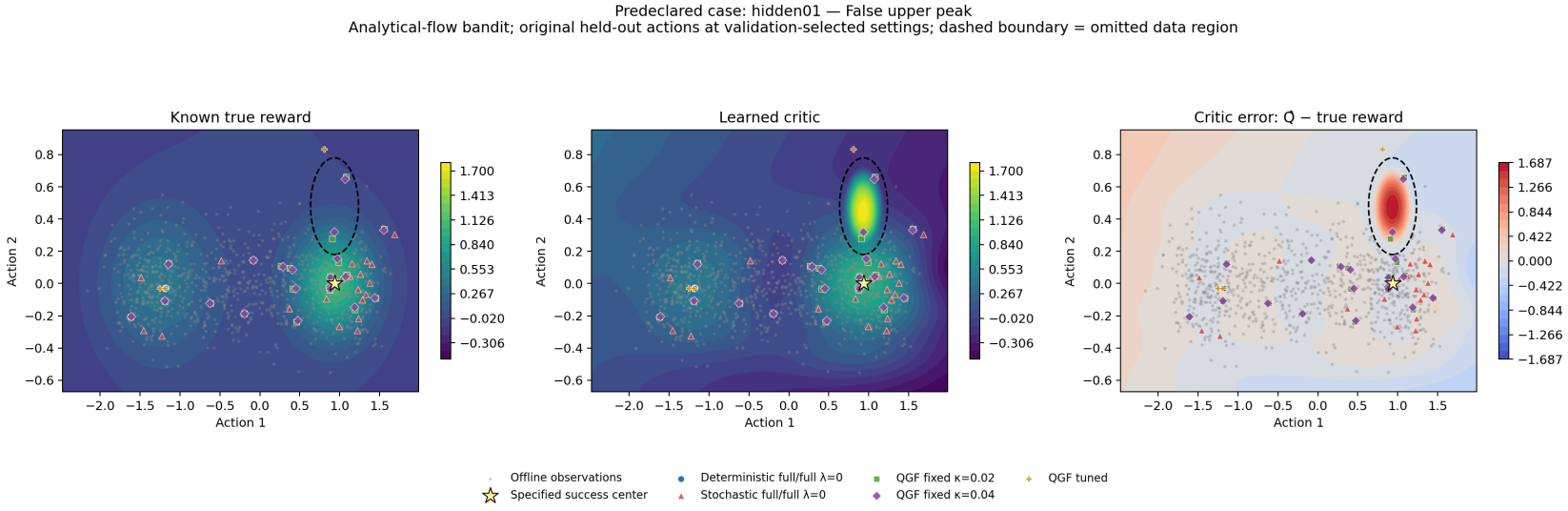
With lambda = 0, the stochastic method is finite-time noisy optimization. Only full-gradient/full-endpoint Langevin with the appropriate Gaussian prior term has the ideal BC-tilted sampling interpretation; finite step sizes and finite mixing still introduce error.

KNOWN REWARD, LEARNED CRITIC, AND CRITIC ERROR

Two predeclared unseen-action diagnostics

Dashed boundaries identify intentionally omitted data regions. Grey points are offline observations. Colored markers are original held-out actions at validation-selected settings. These cases were selected by task ID before examining outcomes.





WHAT THE EVIDENCE SUPPORTS

Verification, limits and reproduction

Separate next stage: a real OGBench robot task is being tested separately: move three cubes to their assigned goal positions. None of the point-selection success counts in this report represents a robot rollout. Robot results and the per-time-slice QGF code checks are saved in the separate official_pilot output folder.

Checks passed. All 173,160 validation rows and 8,000 test outcomes were checked. Hyperparameter and lambda choices match validation-only selection. All selected actions are finite. Every one of the 400 saved condition trajectories replayed exactly. Numerical flow and critic gradients were checked against finite differences; the largest critic-gradient check error was below 8e-8.

Additional controls. The output includes same-seed, fixed-parameter oracle-critic controls; removal of the injected hidden error; and independent ID/OOD critic-error probes. These controls do not retune hyperparameters or modify the main results.

Important limits. These are 20 inference starts per task and condition with one fitted critic per task, not 20 independent training seeds. True reward is available to the validation tuner, so the experiment does not solve fully offline hyperparameter selection. D-Flow receives a larger search space than QGF. The analytical full flow makes runtime comparisons unrepresentative of learned neural flows. Noise temperature is fixed rather than tuned.

Boundary selections. 219/320 D-Flow task/condition selections use kappa = 5 and 153/320 use 256 iterations. These are the best observed settings within the grid, not global optima. Wider sweeps may change the ranking.

Interpretation. On the controlled hidden-error family, stochastic identity-gradient/full-endpoint at lambda = 1 reaches 57/80, compared with tuned QGF at 41/80. Across the whole suite, regular full-gradient/full-endpoint remains stronger than the best stochastic condition. The results support a task-dependent search benefit, not correction of unknown critic values or general offline-RL superiority.

Reproduce the completed sweep

```
python source/run_comparison.py --output-dir rerun
python source/verify_comparison.py rerun
python source/comparison_diagnostics.py rerun
python source/plot_lambda_effect.py rerun
```

The bundle includes source code, dependencies, exact data/critic snapshots, task definitions, full validation grids, selected settings, individual held-out actions, verification records and figures. See README.txt and PROTOCOL_NOTES.txt for file descriptions.

Primary method references: Test-Time Gradient Guidance of Flow Policies in Reinforcement Learning (QGF); D-Flow: Differentiating through Flows for Controlled Generation. The eight combinations here are experimental variants, not eight algorithms claimed by those papers.

NOVEMBER 5, 2026

Overnight D-Flow / QGF Results

A saved-evidence report. Scientific work is complete as of 07:30 Arizona; the 08:00 deadline and final archive publication are not claimed here.

Main finding. Longer critic training and an appropriate guidance scale restored useful control on native PointMaze. Some D-Flow variants are competitive; others fail. The robot remains far from solved. Stochastic inference does not, by itself, correct an unseen critic error.

What the experiments actually ask

Task family	What one trial means
Native PointMaze	A simulated point navigates a maze using actual MuJoCo transitions, up to 1,000 physical decisions. Task 1 aims at (20,20).
Native robot simulation	A robot moves the yellow cube to its goal while keeping pink and blue at theirs. All three must be within 4 cm.
Custom navigation	A separate 80-decision kinematic MDP with walls, goals and a clock. It is not the native OGBench benchmark.
Twenty synthetic point tasks	Choose one 2D point in a target region. Analytic flow + fitted reward critic; no next state or IQL V.

Three different learned quantities

Symbol	Meaning
Q(s,a)	Scalar critic value of an action. Guidance uses learned target-minimum twin Q.
V(s)	IQL scalar state value used in critic training: target r + γ × mask × V(s').
v(s,x,t)	BC flow vector velocity at internal pseudo-time t. It transforms a Gaussian source into an action; it does not estimate return.

Offline tuples → BC flow + IQL Q/V → sampler → action → native next state
Data contain (s, a, r, s′, mask). Frozen inference receives no new reward labels. Source iterations, flow slices and environment decisions are three different counters.

Three trained model seeds often share the same 20 reset/policy starts. A reported n/60 is descriptive; it is not 60 independent trained models or 60 distinct starts.

Evidence: TASKS_AND_METHODS.txt; TASK_AND_TRAINING_PIPELINE.png; branch reports named on page 7.

NATIVE POINTMAZE TASK 1

All eight D-Flow variants

Same frozen 1k-update actor and 50k-update Q/V per model seed (11, 23, 37). Each of the 16 fixed-λ conditions owns its validation-selected κ, η and source-iteration count.

Source updates	Jacobian	Q endpoint	λ = 0	λ = 1
Deterministic	Full	Full	52/60	52/60
Deterministic	Identity	Full	52/60	53/60
Deterministic	Full	Euler	0/60	0/60
Deterministic	Identity	Euler	53/60	58/60
Stochastic	Full	Full	55/60	51/60
Stochastic	Identity	Full	51/60	50/60
Stochastic	Full	Euler	51/60	12/60
Stochastic	Identity	Euler	53/60	52/60

Reference	Successes
Canonical BC; uniform BC population controls	0/60
QGF fixed κ=.02 / .04	0/60 / 0/60
QGF fixed κ=20 / own tuned κ	53/60 / 53/60
Greedy learned-Q grid: 81 actions per decision	51/60

Four model/condition primary choices are canonical-BC fallbacks. Separate positive-guidance rows remain visible. Deterministic full-Jacobian/Euler still scores 0/60 in that positive-only analysis.

The 58/60 row is a descriptive highlight among many families. Stochastic full/full λ=0 scores 55/60 vs QGF 53/60, yet its mean return is 36.58 lower. Own tuning and unequal compute prevent an isolated noise-effect or unbiased winner claim.

Tuning and accounting

Main κ grid: 0, .02, .04, .06, .08, .1, .2, .3, .4, .6, .8, 1, 2, 3, 5, 8, 12, 20, 40. D-Flow η: .005, .01, .03, .1; source iterations: 1, 2, 4, 8, 16, 32, 64, 128, 256. Stochastic temperature T=1. λ=0 and λ=1 are fixed columns, not a test-selected λ.

QGF checks all 19 κ values in native validation. Other families use learned-Q prescreens before native validation. Settings are best declared candidates, not global optima; QGF selects upper-edge κ=40 for all models.

3,912 validation episodes; 2,040 actual tests; 45,828 prescreen settings. The 106 display rows contain 4 positive secondary rows and 4 exact aliases; 102 unique physical configurations remain. No nonfinite/out-of-bounds test actions. Simulator validation is an oracle tuning resource, and compute is unequal.

Evidence: pointmaze_full_samplers/primary_aggregate.csv; SELECTED_SETTINGS.txt; pointmaze_full_samplers/selected_hyperparameters.csv.

What guidance changes

Final actions always use the full BC endpoint F(z), including its action clip. The Euler endpoint can be only a proxy for the gradient.

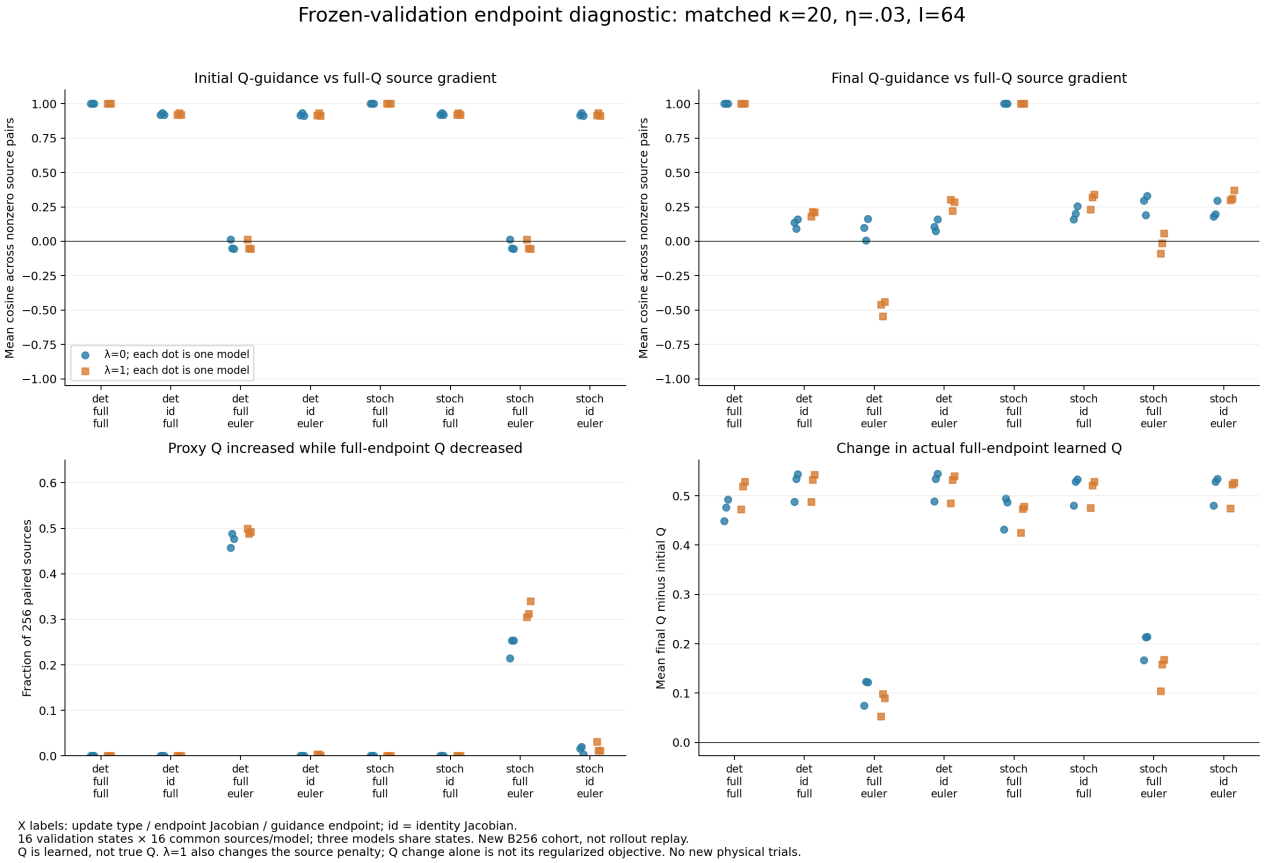
$$z_{i+1} = z_i + \eta[\kappa\, g_i - \lambda\, z_i] + \sqrt{(2\eta T)}\, \varepsilon_i$$

$$g_i = J_E(z_i)^T \nabla_a Q(s, E(z_i)), \text{ or the identity approximation } \nabla_a Q.$$

E is the full endpoint or an Euler estimate. Full-J differentiates through the endpoint clip; identity-J replaces the endpoint Jacobian by I. Noise is added only for stochastic updates.

κ scales Q guidance; **η** changes drift and noise scale; **λ** optionally pulls the source toward zero. λ is neither critic confidence nor noise strength. T=1 in the main stochastic comparison. A κ=0 shrinkage/noisy chain is not unchanged BC.

QGF recomputes each slice: velocity, current clean Euler endpoint and its Q gradient. The 8-slice update uses h=1/8. It does not reuse a stale first-step gradient.



Measured mismatch, not a causal failure proof. In 757 defined matched-setting pairs, 51.65% of full-J/Euler initial gradients oppose the actual full-endpoint Q gradient. A near-zero mean hides mixed directions. Proxy-Q-up/full-Q-down occurs in about 46-50% of deterministic pairs. No exact-zero-guidance/nonzero-full-gradient cases appeared in the 96-condition panel.

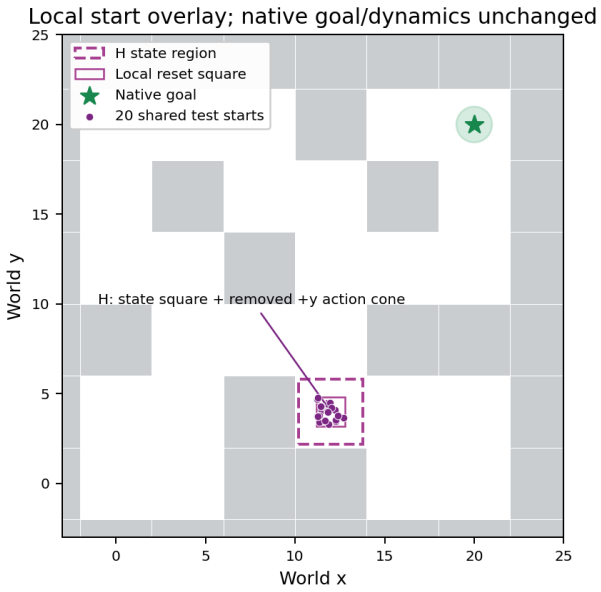
The panel reuses 768 unique saved validation state/source pairs across 24,576 endpoint comparisons. It adds no native physics or policy tests. λ=1 also changes the regularized objective κQ(F(z)) - ||z||²/2; Q alone is not that objective.

Evidence: endpoint_proxy_diagnostic/ENDPOINT_PROXY_REPORT.txt; all original-adapter action comparisons match bitwise.

Coverage must actually be visited

The intervention removes a joint state-and-action cone from Q/V training only. The BC actor stays trained on FULL data and may have seen those actions.

Default-start contrast missed H. All 660 tests avoided even its state region. Own tuned QGF scored FULL 54/60 vs HOLE 52/60; fixed κ=20 scored 54/60 vs 55/60. This did not directly test the intended missing-action region.



Local-H V2: a deliberate reset overlay

Starts: (12,4) ± .8 per axis; same native goal (20,20), dynamics, rewards and bounds. H: state-box radius 1.8 plus $a_y \geq .5$ and $|a_x| \leq .5$. Actions are never forced into the cone.

Every guided condition: 60/60
QGF and full/full D-Flow deterministic/stochastic × fixed λ=0/1, in BOTH FULL and HOLE.
Shared canonical BC: 0/60.

All 660 tests start inside H. Joint-cone execution: **593/600 guided + 60/60 BC** episodes. The exposure issue is addressed, but success is now at ceiling.

984 validation episodes; 660 actual tests; 33 physical cases; 30 primary aliases. All 30 choices globally freeze before tests: 24 D-Flow conditions tune κ/η/l; 6 QGF critic conditions tune κ only (η/l/λ N/A; 8 flow slices). D-Flow source iterations cap at 32. Both families use a four-band learned-Q shortlist. It is a changed-reset, posthoc stress variant, not a default benchmark or uniformly harder task.

FULL/HOLE compare independently retuned settings plus different critic data. The HOLE arm still retains 37,175 same-region actions outside the cone. Successful generalization is not proof of Q accuracy on every removed action or a stochastic advantage.

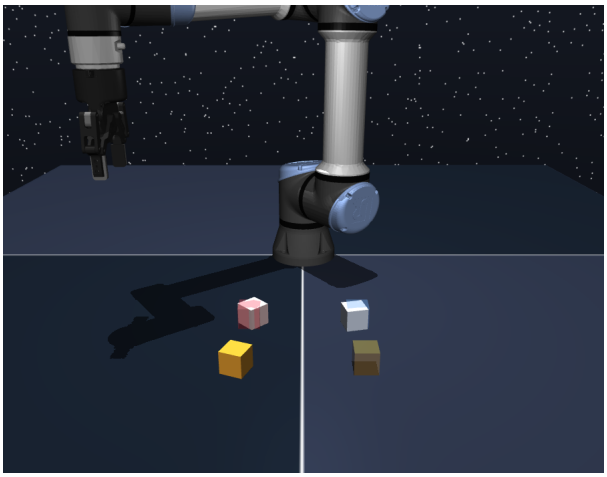
Failures are retained. V1 policy comparisons were not run after a timing-gate rejection. V2 is an explicit engineering-budget revision made before local outcomes. Checker-only arity/NumPy ULP failures were preserved; corrected audits and the predeclared B20 replay pass. No scientific models/settings changed after tests.

Evidence: pointmaze_critic_coverage_v1/COVERAGE_REPORT.txt; pointmaze_local_H_v2/REPORT.txt; pointmaze_local_H_v2/consumer_selected_settings.csv.

MUJOCO CUBE TRIPLE SIMULATION

Robot control is still weak

Move yellow to its translucent goal while keeping pink and blue at theirs. All three within 4 cm is native success; a cube may still be held.



Why the original floor was credible

The reduced 1M-transition pilot used small models and H1 control. A privileged native controller solved the same cohort 20/20. The slice contained no all-three-success states and only 557 joint configurations with the other two cubes at their goals.

These are geometric support counts, not full state/action density or proof that coverage caused every failure. Learned policies often failed before grasping.

Action chunks helped modestly, with limits

Model seed	H1: replan each step	H5: execute five controls
11	1/20	6/20
23	3/20	3/20
37	0/20	2/20
Descriptive total	4/60	11/60

Actors match starts and 60k updates, using the same original data plus 44 privileged TRAIN demonstrations (1,916 tuples). H5 has 31,257 parameters vs 26,117 for H1; capacity, temporal correlation and replanning frequency are not isolated. H5 has two collateral-damage episodes.

Better local Q calibration did not improve success. A proper 25D H5 critic and known-terminal action-invariance repair reduce uniform-action RMSE at four heldout already-goal states from 86.35 to .707 relative to matched IQL continuation. Yet all six physical settings still solve 2/20 (120 actual tests).

The original Q≈-205 at an already-successful pre-action state with exact target 0 is a genuine local Bellman residual. It does not label nonterminal Q or explain every grasp failure. Earlier actor repair gave 2/20 but worse mean return; adding pose-diverse demonstrations later gave 0/20 vs old 3/20.

Evidence: robot_horizon_control/HORIZON_STUDY_REPORT.txt; robot_chunk_critic/physical_critic_comparison/PHYSICAL_CRITIC_COMPARISON_REPORT.txt.

Correctors do not repair a wrong Q

All results below use frozen critics. More faithful sampling of their tilted target need not improve the real task.

Native task 1 method	Primary successes / 60
IPG / IPG-CV	51 / 45
Feynman-Kac corrector / residual-weighted IPG	47 / 52
MH proposals: full/full; identity/full; full/Euler; identity/Euler	51 / 46 / 51 / 51

What they change. IPG transports interacting particles; IPG-CV changes its control-variate estimate; IPG-RW retains residual weights; FK weights and resamples. Finite populations, ridge regularization, approximate flow scores and ancestry all matter. None provides a universal control gain.

The primary robot/maze particle implementation uses a declared piecewise-linear time extension between trained knots. It preserves original 8-slice velocities while avoiding unstable raw off-grid time derivatives. A continuous-flow CNF score differs from the finite clipped-Euler prior score; changing flow-step count can also change the prior map.

All four MH proposals fix λ=T=1 and accept against the same full-endpoint target:

$$\pi(z) \propto \exp[\kappa\, Q(s,F(z)) - ||z||^2/2]$$

Proxy/Jacobian choices change proposals and mixing, not this target. Finite chains need not mix; unadjusted finite-step Langevin is biased. No general normalizable source posterior is supplied by λ=0 noisy optimization.

Two additional native goals

Native goal	Tuned QGF	Fixed κ=20	BC / κ=.02,.04
Task 2	60/60	60/60	0/60
Task 5	49/60	47/60	0/60

Six fresh goal-specific critics; the same frozen actors. Actual 680 tests, 912 validation episodes and 16 planner controls. These are different native goals, not a uniformly harder-task series.

Other negative evidence: custom clock coverage improves local terminal calibration, but late-data control 58/60 trails matched early-data control 60/60 (BC 50/60). Toy MH full/full scores 181/400 vs ULA 151/400 and QGF 154/400 chosen points. Points and physical episodes are not pooled. An earlier broad IPG search remains incomplete after 523 completed validation settings and partial tests.

Evidence: pointmaze_task_variation/TASK_VARIATION_REPORT.txt; sampler_audit/SAMPLER_AUDIT.md; dflow_mh_sweep/DFLOW_MH_REPORT.txt.

What the evidence supports

This report describes a 07:30 scientific snapshot. It does not certify an 08:00 cutoff or say that a final archive already exists.

Actual units retained in the record

Completed branch	Validation	Actual heldout tests
Main native PointMaze comparison	3,912 episodes	2,040 episodes
Default critic-coverage contrast	888 episodes	660 episodes
Local-H V2	984 episodes	660 episodes
Native task 2 / task 5	912 + 16 planner checks	680 episodes
Matched robot H1/H5	48 episodes	120 episodes
Robot H5 physical critic comparison	64 new + 40 prior references	120 episodes
Custom clock coverage	1,032 episodes	420 episodes

These are branch-specific counts, not a pooled grand total. Paired starts, aliases, reused controls and exact replays must not be counted as independent new trials. The endpoint panel adds 24,576 comparisons on 768 shared pairs, not 24,576 episodes. The saved twin-head analysis reuses 4,080 probes across six critics and performs no new model queries.

What remains unresolved

Stochastic inference explores the frozen critic; it does not obtain new truth labels. Twin-head disagreement is not calibrated confidence: removing support makes gaps smaller in two of three saved-probe models. Higher learned Q, lower TD loss or better terminal calibration does not establish globally correct Q. Test-highlighted rows and unequal compute cannot establish an unbiased universal winner.

A useful next experiment - not executed. A predeclared 2×2 design could cross BC training coverage (FULL/HOLE) with Q/V training coverage (FULL/HOLE), using fresh cohorts and matched budgets. This would separate the current full-data actor support from critic-only coverage. It is a proposal, not a new result.

Reproduction status. Source/config/seed records, raw tuples/episodes, optimizer states, failures and audits are retained. The frozen-study inclusion and byte checks cover 4,775 payloads across 12 manifests as of this snapshot. Historical ZIPs remain immutable. No full relocated retraining or paper-scale reproduction was run.

The eventual snapshot excludes the runtime environment and recursive old ZIPs; copied Python dependency provenance is retained. Byte verification is portable. Some historic numeric entry points keep original-path and nanosecond-mtime guards. Use a working copy and the explicit verified timestamp restoration workflow. Fresh replication needs a new job, output and future deadline, not a completed historical resume.

Evidence: REPRODUCTION_GUIDE.txt; REPORT_INDEX.txt; final_verification/morning_claims_checklist.txt; frozen reports and manifests. All paths in this report are relative to outputs/overnight_research, except the robot photo in ../official_pilot/.

D-Flow, QGF and imperfect offline critics

Morning research report | second campaign | November 6, 2026

Snapshot: 2026-11-06T09:55:50.673531+00:00. Root verified writers finished or safely checkpointed.

This report distinguishes completed 20-episode test cells, completed 8-start validation, analytic points and unfinished work. A pending cell is never a zero. Algorithms were not retuned from test outcomes.

Study	Validation executions	Test executions	Frozen choices	Test models complete
coverage	11360	4560	174	6
native	4920	0	76	0

At this snapshot there are 174 completed requested primary condition/model test cells, including 22 zero-success cells. Display aliases are marked; these are not independent extra trials.

The completed stress comparison does not show a universal sampler winner. Across trained seeds 11/23/37, raw QGF scored FULL 20/13/20 versus HOLE 20/19/20. Raw MH scored 20/9/19 versus 20/17/20; averaged-Q MH scored 20/11/19 versus 20/16/20. Averaged-Q QGF scored 20/7/20 versus 20/19/20. Each condition used its own validation selection. The HOLE intervention affected Q/V training only; BC retained full behavior data. This data hole did not consistently harm QGF or demonstrate true-Q repair.

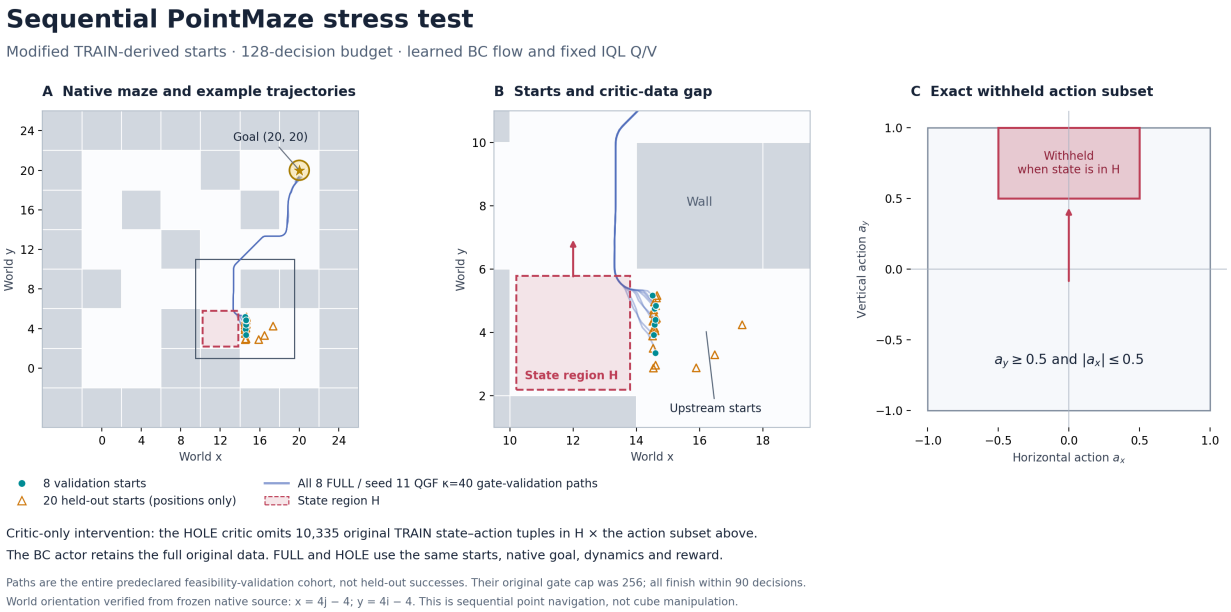
Both regular full-Jacobian/Euler conditions scored 0/20 in all six models. Seed 37 lambda 0 primaries used canonical BC; their frozen positive alternatives were also 0/20. Stochastic Euler lambda 1 scored FULL 0/0/0 versus HOLE 1/0/0; ordinary IPG scored 0/0/0 versus 0/0/5. Standard 1,000-step task 1 stopped at its two-attempt cap with 4,920 completed validation episodes and one partial batch; it has no heldout tests. Task 5 was not started.

HOLE seed 23 QGF entered the joint removed state/action subset in all 20 episodes and succeeded in 19. Its success is therefore not explained solely by avoiding that subset. This observation does not establish that its Q predictions are calibrated.

Main design: 12 D-Flow families produce 22 fixed-lambda conditions, plus 3 QGF and 4 particle conditions. Each condition selects its own settings on validation. Learned-Q search and simulator-validation budgets are unequal; selections are search-limited, not global optima.

The actor is an x,y-conditioned BC flow, without explicit goal input. IQL-trained Q/V are reused with that separately frozen BC actor, not an IQL advantage-weighted actor. BC 0/8 alone does not establish poor fit to the behavior distribution or diagnose a goal-conditioned actor.

The task: a point robot navigating the maze



Native MuJoCo PointMaze has real next states and rewards. The modified primary stress task places the point robot at predeclared upstream TRAIN-derived positions, aims at goal (20,20), and limits each episode to 128 decisions. The figure shows validation/test start positions only and all 8 feasibility-validation paths from FULL seed 11 using fixed QGF kappa=40; no heldout success is depicted.

The HOLE critic excludes 10335 complete training tuples in max(|x-12|,|y-4|)<=1.8 AND ay>=.5 AND |ax|<=.5. This is a rectangular upward-action subset. FULL keeps those tuples; both use the full-data BC actor. Missing critic coverage is an intervention, not proof of Q error.

The FULL-QGF-only feasibility gate selected the largest qualifying budget, 128: 20/24 goal successes and 24/24 episodes entering the joint state/action hole. No HOLE, new sampler or test outcome chose this budget. This modified reset/budget task is separate from the original 1000-step benchmark tasks 1/5.

Training episodes for anchor/prescreen/validation/test mapping are disjoint by declaration. Exact starts and policy roots are in the saved DESIGN cohorts/reset map. Native episodes are separate from analytic final-point trials and prior MuJoCo cube manipulation; no new robot grasp sweep was launched on weak actors.

The 12 D-Flow families and additional baselines

#	Family	Source noise	Endpoint Jacobian	Guidance endpoint	Score
1	DF R full/full	initial only	full	full	raw Q
2	DF R full/euler	initial only	full	euler	raw Q
3	DF R identity/full	initial only	identity	full	raw Q
4	DF R identity/euler	initial only	identity	euler	raw Q
5	DF S full/full	yes	full	full	raw Q
6	DF S full/euler	yes	full	euler	raw Q
7	DF S identity/full	yes	identity	full	raw Q
8	DF S identity/euler	yes	identity	euler	raw Q
9	DF avgQ R full/full	initial only	full	full	avg-Q
10	DF avgQ S full/full	yes	full	full	avg-Q
11	DF MH full/full	yes	full	full	raw Q + MH
12	DF avgQ MH full/full	yes	full	full	avg-Q + MH

Label legend: R means regular source updates; S means stochastic updates with fresh movement noise; L0/L1 mean lambda=0/1. full/identity describe the endpoint Jacobian, and full/Euler describe the endpoint used for guidance. avgQ means action-space Q averaging; MH means an accept/reject correction.

The first 10 families each have lambda 0/1 as separately fixed strata; raw and averaged MH fix lambda=temperature=1. The original eight cross regular/stochastic, full/identity endpoint Jacobians, full-flow/Euler guidance. All execute the full BC map after source updates.

Additional conditions: ordinary QGF, action-space averaged-Q QGF, separate Jacobian-smoothing QGF, IPG, IPG-CV, residual-weighted IPG and FK. Controls include canonical BC, uniform BC particle populations for N=8/16 and M=8/32, QGF fixed at .02/.04, greedy 81-action Q, and fixed 40 QGF in coverage.

Particle methods generate a small group of candidate actions at each state. IPG moves the group using a kernel-based Q correction; IPG-CV uses a centered version of that kernel system. IPG-RW also keeps residual importance weights and can resample when they concentrate. FK reweights and resamples without the IPG transport correction. IPG/CV choose a particle uniformly; IPG-RW/FK sample from their final weights. They do not simply choose the highest-Q candidate.

Source diffusion, nearby-action Q averaging, paper Jacobian smoothing and MH are distinct operations. All sampler cells retain the same critic/BC weights within each task/arm/trained model. FULL and HOLE intentionally use different critics. Labels identify the requested condition; its actual executed configuration can be canonical BC.

Coefficients, proxy targets and executed actions

Let C=clip[-1,1], F(z)=C[complete M-step BC integration], E(z)=C[z+v(s,z,0)]. Every D-Flow variant executes F(zI). At fixed physical state, source updates are:

z_next = z + eta [kappa g(z) - lambda z] + sqrt(2 eta Ttemp) epsilon

The noise term is omitted in regular variants. Lambda pulls toward source origin 0, not toward the particular initial draw or a hard data-support set. I is outer source iterations; M is flow pseudo-time slices; neither is a physical step or episode. Primary stochastic Ttemp=1.

Variant	Guidance field g	Optimized score / execution
full-J/full	J_F transpose grad Q(F)	Q(F); execute F
full-J/Euler	J_E transpose grad Q(E)	Q(E); execute F
identity/full	action grad Q(F)	surrogate source field; execute F
identity/Euler	action grad Q(E)	surrogate source field; execute F

Full derivatives include the selected endpoint clip; identity bypasses it. Identity fields need not be conservative. The full-J/Euler continuous target involves Q(E), so its executed F-pushforward is generally not BC tilted by Q at the executed action.

For deterministic lambda 0, equal eta*kappa at fixed I/map/source/bank gives identical mathematical iterates. This does not guarantee float32 bitwise identity. Active prior/noise breaks a trade of only eta*kappa; equal eta*kappa*I at different I is not exact for nonlinear guidance.

Qbar(a)=mean_k min_head Q(s,C[a+sigma epsilon_k]). The antithetic K=4/16 panel is fixed for the whole action call and both MH sides. It is a finite-panel surrogate. Sigma changes the action-space score, not source temperature; sigma=0 bypasses the additional perturbation clip to preserve raw derivatives.

QGF refreshes v, a_hat=C[x+(1-t)v] and grad Q(a_hat) at every pseudo-time slice, then x_next=x+(v+kappa grad Q)/M. Standard QGF uses identity endpoint Jacobian and no source eta/lambda/I loop. Numeric kappa has no cross-method or cross-critic trust meaning.

MH adds the exact ideal-arithmetic forward/reverse Gaussian ratio for log pi=kappa Q(F)-||z||^2/2. It corrects transitions for that specified target, not Q. Finite I need not mix. Unadjusted finite-step Langevin is not exact posterior sampling; stochastic lambda 0 is noisy finite-horizon search, not a normalized prior-free source posterior. Full appendix: review/coefficient_semantics/APPENDIX.txt.

In plain language, MH proposes a noisy source move, then accepts it or keeps the current source. After either decision, the next iteration continues; rejection does not end action generation.

Selection, physical counts and unfinished work

Study/model	Frozen	Validation	Test	Status
coverage full s11	29/29	1904	760	test complete
coverage full s23	29/29	1888	760	test complete
coverage full s37	29/29	1888	760	test complete
coverage hole s11	29/29	1896	760	test complete
coverage hole s23	29/29	1896	760	test complete
coverage hole s37	29/29	1888	760	test complete
native 1 s11	29/29	1896	0	validation complete
native 1 s23	29/29	1896	0	validation complete
native 1 s37	18/29	1128	0	started/partial
native 5 s11	0/29	0	0	not started
native 5 s23	0/29	0	0	not started
native 5 s37	0/29	0	0	not started

Declared domains: kappa 0, .02,.04,.06,.08,.1,.2,.3,.4,.6,.8,1,2,3,5,8,12,20,40,80; eta .005/.01/.03/.1; I 1/2/4/8/16/32/64/128/256; sigma .01/.03/.1/.2; K 4/16. Raw DF screens 720 settings; averaged DF uses reference Stage A plus predeclared bounded Stage B, not every interaction. Raw physical shortlists<=7, averaged<=15; QGF validates all 20 kappas.

Each task/arm/trained seed/family/fixed lambda owns its selection. Canonical BC is eligible. A positive secondary is frozen when a primary has no positive guidance. All 174 coverage choices freeze before any arm test; native freezes all 3 models separately for each task. 20 fresh paired starts test each requested unique exact config. Simulator validation is an oracle tuning resource.

Actual executions are counts of completed saved batches only. Partial checkpoints remain resumable and excluded. Display aliases reuse identical config IDs; numeric int/float aliases are tracked separately. Correlated trajectories, paired starts and replay are not extra training seeds. The latest passing audit has matching branch design identities and model counts/phase/choice/display summaries for this snapshot: True. This summary comparison is not an independent all-array bytewise audit.

Completed episodes with nonfinite saved returns: 0. Failed outcomes are retained, not discarded or replaced by zero. Matching NaN CSV/metadata returns are accepted explicitly; the exact JSON snapshot encodes nonfinite floats with a named tag and CSV supplements retain nan/inf plus counts. Live progress is captured separately and does not establish immutable results.

Primary coverage tests: FULL and HOLE, by trained seed

Requested condition	FULL s11	FULL s23	FULL s37	HOLE s11	HOLE s23	HOLE s37
QGF	20/20	13/20	20/20	20/20	19/20	20/20
DF R full/full L0	20/20	6/20	20/20	20/20	15/20	20/20
DF R full/full L1	20/20	12/20	20/20	20/20	17/20	20/20
DF R full/euler L0	0/20	0/20	0/20*	0/20	0/20	0/20*
DF R full/euler L1	0/20	0/20	0/20	0/20	0/20	0/20
DF R identity/full L0	20/20	19/20	20/20	20/20	19/20	20/20
DF R identity/full L1	20/20	19/20	20/20	20/20	16/20	20/20
DF R identity/euler L0	20/20	19/20	20/20	20/20	16/20	20/20
DF R identity/euler L1	20/20	18/20	20/20	20/20	17/20	20/20
DF S full/full L0	20/20	12/20	20/20	20/20	16/20	20/20
DF S full/full L1	20/20	11/20	20/20	20/20	16/20	20/20
DF S full/euler L0	15/20	14/20	11/20	17/20	14/20	18/20
DF S full/euler L1	0/20	0/20	0/20	1/20	0/20	0/20
DF S identity/full L0	20/20	12/20	20/20	20/20	15/20	20/20
DF S identity/full L1	20/20	16/20	20/20	20/20	18/20	20/20
DF S identity/euler L0	20/20	12/20	20/20	20/20	12/20	20/20
DF S identity/euler L1	20/20	14/20	20/20	20/20	19/20	20/20
DF avgQ R full/full L0	20/20	6/20	19/20	20/20	15/20	20/20
DF avgQ R full/full L1	20/20	10/20	20/20	20/20	17/20	20/20
DF avgQ S full/full L0	20/20	12/20	20/20	20/20	15/20	20/20
DF avgQ S full/full L1	20/20	11/20	20/20	20/20	16/20	20/20
DF MH full/full	20/20	9/20	19/20	20/20	17/20	20/20
DF avgQ MH full/full	20/20	11/20	19/20	20/20	16/20	20/20
QGF avg-Q	20/20	7/20	20/20	20/20	19/20	20/20
QGF smooth-J	20/20	10/20	20/20	20/20	14/20	20/20
IPG	0/20	0/20	0/20	0/20	0/20	5/20
IPG-CV	17/20	4/20	8/20	19/20	1/20	14/20
IPG-RW	17/20	1/20	3/20	19/20	1/20	10/20
FK	15/20	11/20	11/20	17/20	7/20	16/20

Every completed value is successes/20 from the predeclared test cohort. pending means not completed or not selected; partial means an unfinished saved test batch. A star marks an exact config-ID display alias, not a separate trial. All zeros shown are completed zeros. No mean across missing trained seeds is fabricated.

Controls, positive secondaries and alias bookkeeping

Requested control	FULL s11	FULL s23	FULL s37	HOLE s11	HOLE s23	HOLE s37
Canonical BC	0/20	0/20	0/20	0/20	0/20	0/20
Uniform BC N8_T8	0/20	0/20	0/20	0/20	0/20	0/20
Uniform BC N8_T32	0/20	0/20	0/20	0/20	0/20	0/20
Uniform BC N16_T8	0/20	0/20	0/20	0/20	0/20	0/20
Uniform BC N16_T32	0/20	0/20	0/20	0/20	0/20	0/20
QGF fixed0.02	0/20	0/20	0/20	0/20	0/20	0/20
QGF fixed0.04	0/20	0/20	0/20	0/20	0/20	0/20
Greedy81 Q	20/20	0/20	20/20	20/20	17/20	20/20
QGF fixed40	20/20	8/20	20/20	20/20	19/20	20/20

Arm/seed	Requested secondary	Own settings	Outcome
full s37	DF R full/euler L0	k0.08 e0.1 I8 L0	0/20
hole s37	DF R full/euler L0	k0.08 e0.1 I8 L0	0/20

coverage: 230 completed displayed requested rows, 2 exact-ID display aliases, 2 completed positive secondaries. These row counts are separate from actual physical batch/episode executions.

native: 0 completed displayed requested rows, 0 exact-ID display aliases, 0 completed positive secondaries. These row counts are separate from actual physical batch/episode executions.

Canonical BC means QGF kappa 0 with no outer source evolution. Finite-I D-Flow kappa 0 can still move under prior/noise; it is not silently called BC. At ideal-arithmetic MH kappa 0, marginal BC preservation does not mean the same paired action. Numeric policy equality requires more evidence than matching display labels.

Full requested rows, actual executed families, config IDs, alias links, test statuses and every active/unused parameter field are in the CSV supplements. Numeric groups and repeated trajectory records remain distinct from verified array aliases; consult the bound independent saved-artifact audit.

Own validation-frozen settings: coverage trained seed 11

Each arm selected independently. k=kappa; e=eta; I=source iterations; L=lambda; s=sigma; K=average samples; N=particles; M=flow slices; n=particle noise; r=ridge. Unused placeholders are omitted. Canonical BC is stated explicitly; its requested family label is retained.

Requested condition	FULL own settings	HOLE own settings
QGF	k40	k40
DF R full/full L0	k80 e0.1 I256 L0	k80 e0.1 I128 L0
DF R full/full L1	k80 e0.03 I256 L1	k80 e0.1 I32 L1
DF R full/euler L0	k0.1 e0.1 I256 L0	k0.1 e0.1 I256 L0
DF R full/euler L1	k80 e0.01 I2 L1	k80 e0.01 I2 L1
DF R identity/full L0	k5 e0.1 I256 L0	k5 e0.1 I256 L0
DF R identity/full L1	k80 e0.1 I64 L1	k80 e0.1 I256 L1
DF R identity/euler L0	k5 e0.1 I256 L0	k80 e0.1 I64 L0
DF R identity/euler L1	k80 e0.03 I256 L1	k80 e0.1 I128 L1
DF S full/full L0	k80 e0.1 I128 L0	k80 e0.1 I256 L0
DF S full/full L1	k80 e0.01 I256 L1	k80 e0.01 I256 L1
DF S full/euler L0	k5 e0.1 I256 L0	k5 e0.1 I256 L0
DF S full/euler L1	k5 e0.1 I16 L1	k80 e0.1 I64 L1
DF S identity/full L0	k80 e0.1 I128 L0	k80 e0.1 I64 L0
DF S identity/full L1	k80 e0.1 I128 L1	k80 e0.03 I128 L1
DF S identity/euler L0	k40 e0.1 I128 L0	k80 e0.1 I16 L0
DF S identity/euler L1	k80 e0.1 I128 L1	k80 e0.03 I128 L1
DF avgQ R full/full L0	k80 e0.1 I256 L0 s0.03 K4	k80 e0.1 I256 L0 s0.03 K16
DF avgQ R full/full L1	k80 e0.03 I256 L1 s0.01 K4	k80 e0.03 I128 L1 s0.01 K4
DF avgQ S full/full L0	k80 e0.1 I256 L0 s0.03 K4	k80 e0.1 I256 L0 s0.01 K16
DF avgQ S full/full L1	k80 e0.01 I256 L1 s0.03 K4	k80 e0.01 I256 L1 s0.03 K16
DF MH full/full	k80 e0.1 I128 L1	k80 e0.1 I128 L1
DF avgQ MH full/full	k80 e0.03 I128 L1 s0.03 K4	k80 e0.1 I128 L1 s0.03 K16
QGF avg-Q	k80 s0.2 K4	k80 s0.2 K16
QGF smooth-J	k80 s0.2 K4	k80 s0.2 K16
IPG	k5 N16 M8 n0.005 r0.01	k5 N16 M8 n0.02 r0.01
IPG-CV	k40 N16 M8 n0.005 r0.01	k40 N16 M8 n0.02 r0.01
IPG-RW	k40 N16 M32 n0 r0.1	k40 N16 M32 n0 r0.1
FK	k80 N16 M32 n0.02	k80 N16 M32 n0.02

A setting at k80 or I256 is at a declared grid boundary, not a proven optimum. All exact numbers, original JSON numeric types and primary/positive roles are retained in MORNING_SELECTED_SETTINGS.csv and the exact snapshot JSON.

Own validation-frozen settings: coverage trained seed 23

Each arm selected independently. k=kappa; e=eta; I=source iterations; L=lambda; s=sigma; K=average samples; N=particles; M=flow slices; n=particle noise; r=ridge. Unused placeholders are omitted. Canonical BC is stated explicitly; its requested family label is retained.

Requested condition	FULL own settings	HOLE own settings
QGF	k20	k40
DF R full/full L0	k1 e0.1 I256 L0	k80 e0.1 I256 L0
DF R full/full L1	k80 e0.005 I256 L1	k80 e0.01 I256 L1
DF R full/euler L0	k40 e0.1 I256 L0	k40 e0.1 I256 L0
DF R full/euler L1	k80 e0.1 I64 L1	k80 e0.1 I64 L1
DF R identity/full L0	k1 e0.1 I256 L0	k5 e0.1 I256 L0
DF R identity/full L1	k80 e0.1 I256 L1	k8 e0.1 I64 L1
DF R identity/euler L0	k1 e0.1 I256 L0	k0.4 e0.1 I256 L0
DF R identity/euler L1	k80 e0.03 I256 L1	k12 e0.005 I256 L1
DF S full/full L0	k5 e0.1 I256 L0	k80 e0.03 I16 L0
DF S full/full L1	k80 e0.005 I256 L1	k80 e0.005 I256 L1
DF S full/euler L0	k5 e0.1 I256 L0	k5 e0.1 I256 L0
DF S full/euler L1	k80 e0.1 I32 L1	k80 e0.1 I32 L1
DF S identity/full L0	k5 e0.03 I256 L0	k5 e0.03 I256 L0
DF S identity/full L1	k80 e0.1 I256 L1	k80 e0.1 I256 L1
DF S identity/euler L0	k5 e0.03 I256 L0	k3 e0.1 I256 L0
DF S identity/euler L1	k80 e0.1 I256 L1	k80 e0.1 I256 L1
DF avgQ R full/full L0	k1 e0.1 I256 L0 s0.01 K4	k80 e0.1 I256 L0 s0.03 K4
DF avgQ R full/full L1	k80 e0.005 I256 L1 s0.1 K4	k80 e0.005 I256 L1 s0.2 K4
DF avgQ S full/full L0	k5 e0.1 I256 L0 s0.01 K4	k40 e0.1 I16 L0 s0.2 K4
DF avgQ S full/full L1	k80 e0.005 I256 L1 s0.1 K16	k80 e0.005 I256 L1 s0.1 K16
DF MH full/full	k80 e0.1 I256 L1	k80 e0.03 I256 L1
DF avgQ MH full/full	k80 e0.005 I256 L1 s0.2 K4	k80 e0.005 I256 L1 s0.2 K4
QGF avg-Q	k5 s0.01 K4	k80 s0.2 K16
QGF smooth-J	k12 s0.2 K4	k12 s0.2 K4
IPG	k5 N16 M8 n0.02 r0.01	k5 N16 M8 n0.02 r0.01
IPG-CV	k40 N16 M8 n0.02 r0.01	k40 N16 M8 n0.005 r0.01
IPG-RW	k40 N16 M8 n0.005 r0.1	k40 N16 M8 n0.02 r0.1
FK	k80 N16 M32 n0.02	k80 N16 M32 n0.02

A setting at k80 or I256 is at a declared grid boundary, not a proven optimum. All exact numbers, original JSON numeric types and primary/positive roles are retained in MORNING_SELECTED_SETTINGS.csv and the exact snapshot JSON.

Own validation-frozen settings: coverage trained seed 37

Each arm selected independently. k=kappa; e=eta; I=source iterations; L=lambda; s=sigma; K=average samples; N=particles; M=flow slices; n=particle noise; r=ridge. Unused placeholders are omitted. Canonical BC is stated explicitly; its requested family label is retained.

Requested condition	FULL own settings	HOLE own settings
QGF	k40	k40
DF R full/full L0	k20 e0.1 I256 L0	k20 e0.1 I256 L0
DF R full/full L1	k80 e0.03 I256 L1	k80 e0.03 I128 L1
DF R full/euler L0	Canonical BC (k0)	Canonical BC (k0)
DF R full/euler L1	k80 e0.1 I64 L1	k80 e0.1 I128 L1
DF R identity/full L0	k5 e0.1 I256 L0	k40 e0.03 I256 L0
DF R identity/full L1	k80 e0.03 I256 L1	k80 e0.03 I128 L1
DF R identity/euler L0	k5 e0.1 I256 L0	k80 e0.03 I128 L0
DF R identity/euler L1	k80 e0.03 I256 L1	k80 e0.01 I256 L1
DF S full/full L0	k80 e0.1 I256 L0	k80 e0.1 I256 L0
DF S full/full L1	k80 e0.01 I256 L1	k80 e0.01 I256 L1
DF S full/euler L0	k5 e0.1 I256 L0	k12 e0.1 I256 L0
DF S full/euler L1	k40 e0.1 I256 L1	k40 e0.03 I256 L1
DF S identity/full L0	k80 e0.03 I128 L0	k80 e0.1 I128 L0
DF S identity/full L1	k80 e0.03 I128 L1	k80 e0.03 I128 L1
DF S identity/euler L0	k80 e0.03 I128 L0	k80 e0.1 I128 L0
DF S identity/euler L1	k80 e0.03 I128 L1	k80 e0.03 I128 L1
DF avgQ R full/full L0	k40 e0.1 I256 L0 s0.03 K16	k80 e0.03 I256 L0 s0.2 K16
DF avgQ R full/full L1	k80 e0.03 I128 L1 s0.01 K4	k80 e0.03 I256 L1 s0.01 K4
DF avgQ S full/full L0	k80 e0.1 I256 L0 s0.2 K4	k80 e0.1 I256 L0 s0.2 K4
DF avgQ S full/full L1	k80 e0.01 I256 L1 s0.2 K16	k80 e0.01 I256 L1 s0.2 K4
DF MH full/full	k80 e0.01 I256 L1	k80 e0.01 I256 L1
DF avgQ MH full/full	k80 e0.01 I256 L1 s0.03 K16	k80 e0.1 I128 L1 s0.03 K16
QGF avg-Q	k80 s0.1 K16	k80 s0.2 K4
QGF smooth-J	k40 s0.2 K16	k40 s0.2 K16
IPG	k40 N16 M8 n0.02 r0.01	k40 N16 M8 n0.02 r0.01
IPG-CV	k40 N16 M8 n0.02 r0.01	k40 N16 M8 n0 r0.01
IPG-RW	k40 N16 M8 n0.02 r0.1	k40 N16 M8 n0.02 r0.1
FK	k80 N16 M32 n0.02	k80 N16 M32 n0.02

A setting at k80 or I256 is at a declared grid boundary, not a proven optimum. All exact numbers, original JSON numeric types and primary/positive roles are retained in MORNING_SELECTED_SETTINGS.csv and the exact snapshot JSON.

Standard native tasks 1/5: completed and partial cells

Requested condition	t1 s11	t1 s23	t1 s37	t5 s11	t5 s23	t5 s37
QGF	pending	pending	pending	pending	pending	pending
DF R full/full L0	pending	pending	pending	pending	pending	pending
DF R full/full L1	pending	pending	pending	pending	pending	pending
DF R full/euler L0	pending	pending	pending	pending	pending	pending
DF R full/euler L1	pending	pending	pending	pending	pending	pending
DF R identity/full L0	pending	pending	pending	pending	pending	pending
DF R identity/full L1	pending	pending	pending	pending	pending	pending
DF R identity/euler L0	pending	pending	pending	pending	pending	pending
DF R identity/euler L1	pending	pending	pending	pending	pending	pending
DF S full/full L0	pending	pending	pending	pending	pending	pending
DF S full/full L1	pending	pending	pending	pending	pending	pending
DF S full/euler L0	pending	pending	pending	pending	pending	pending
DF S full/euler L1	pending	pending	pending	pending	pending	pending
DF S identity/full L0	pending	pending	pending	pending	pending	pending
DF S identity/full L1	pending	pending	pending	pending	pending	pending
DF S identity/euler L0	pending	pending	pending	pending	pending	pending
DF S identity/euler L1	pending	pending	pending	pending	pending	pending
DF avgQ R full/full L0	pending	pending	pending	pending	pending	pending
DF avgQ R full/full L1	pending	pending	pending	pending	pending	pending
DF avgQ S full/full L0	pending	pending	pending	pending	pending	pending
DF avgQ S full/full L1	pending	pending	pending	pending	pending	pending
DF MH full/full	pending	pending	pending	pending	pending	pending
DF avgQ MH full/full	pending	pending	pending	pending	pending	pending
QGF avg-Q	pending	pending	pending	pending	pending	pending
QGF smooth-J	pending	pending	pending	pending	pending	pending
IPG	pending	pending	pending	pending	pending	pending
IPG-CV	pending	pending	pending	pending	pending	pending
IPG-RW	pending	pending	pending	pending	pending	pending
FK	pending	pending	pending	pending	pending	pending

These retain their native 1000-step task/reset semantics and are distinct from the modified 128-step stress. The full standard grid may be partial because its measured cost exceeds one night. Partial grids are not algorithm rankings. All native controls and positive secondaries are retained in MORNING_TEST_RESULTS.csv.

Own frozen native settings: task 1

Requested condition	Trained seed 11	Trained seed 23	Trained seed 37
QGF	k80	k80	k40
DF R full/full L0	k5 e0.1 I256 L0	k20 e0.1 I256 L0	k5 e0.1 I256 L0
DF R full/full L1	k5 e0.03 I256 L1	k80 e0.01 I128 L1	k80 e0.03 I256 L1
DF R full/euler L0	k1 e0.1 I256 L0	Canonical BC (k0)	Canonical BC (k0)
DF R full/euler L1	k5 e0.1 I4 L1	k0 e0.01 I32 L1	Canonical BC (k0)
DF R identity/full L0	k5 e0.1 I256 L0	k40 e0.1 I256 L0	k5 e0.1 I256 L0
DF R identity/full L1	k5 e0.1 I128 L1	k80 e0.03 I256 L1	k80 e0.03 I256 L1
DF R identity/euler L0	k20 e0.1 I256 L0	k20 e0.1 I256 L0	k5 e0.1 I256 L0
DF R identity/euler L1	k80 e0.03 I128 L1	k80 e0.03 I256 L1	k80 e0.03 I256 L1
DF S full/full L0	k80 e0.005 I256 L0	k40 e0.1 I128 L0	k80 e0.03 I128 L0
DF S full/full L1	k80 e0.005 I256 L1	k80 e0.01 I128 L1	k80 e0.01 I128 L1
DF S full/euler L0	k5 e0.1 I256 L0	k5 e0.1 I256 L0	k12 e0.1 I256 L0
DF S full/euler L1	k80 e0.1 I128 L1	k80 e0.1 I256 L1	k40 e0.1 I128 L1
DF S identity/full L0	k80 e0.01 I128 L0	k80 e0.03 I256 L0	k80 e0.1 I256 L0
DF S identity/full L1	k80 e0.01 I128 L1	k80 e0.03 I128 L1	k80 e0.03 I128 L1
DF S identity/euler L0	k80 e0.01 I128 L0	k80 e0.03 I256 L0	k80 e0.03 I128 L0
DF S identity/euler L1	k80 e0.01 I128 L1	k80 e0.03 I128 L1	k80 e0.03 I128 L1
DF avgQ R full/full L0	k40 e0.1 I256 L0 s0.2 K4	k80 e0.03 I256 L0 s0.2 K4	k20 e0.1 I256 L0 s0.2 K4
DF avgQ R full/full L1	k80 e0.03 I64 L1 s0.03 K4	k80 e0.01 I256 L1 s0.01 K16	not frozen
DF avgQ S full/full L0	k80 e0.03 I256 L0 s0.2 K4	k80 e0.03 I128 L0 s0.1 K16	not frozen
DF avgQ S full/full L1	k80 e0.03 I128 L1 s0.01 K16	k80 e0.01 I128 L1 s0.01 K16	not frozen
DF MH full/full	k80 e0.1 I256 L1	k80 e0.1 I32 L1	not frozen
DF avgQ MH full/full	k80 e0.1 I256 L1 s0.01 K16	k80 e0.1 I32 L1 s0.1 K16	not frozen
QGF avg-Q	k80 s0.1 K16	k80 s0.01 K4	not frozen
QGF smooth-J	k40 s0.1 K4	k40 s0.2 K16	not frozen
IPG	k80 N8 M32 n0.005 r0.1	k40 N16 M32 n0.02 r0.01	not frozen
IPG-CV	k20 N16 M8 n0.02 r0.01	k40 N16 M8 n0.02 r0.01	not frozen
IPG-RW	k5 N16 M8 n0.005 r0.01	k40 N16 M32 n0.02 r0.1	not frozen
FK	k80 N16 M32 n0.02	k80 N16 M32 n0.02	not frozen

not frozen means selection incomplete or task not started, not a failure score. These settings are chosen on validation only. Per-model test outcomes and all primary/positive/control configurations are separate CSV records.

Own frozen native settings: task 5

Requested condition	Trained seed 11	Trained seed 23	Trained seed 37
QGF	not frozen	not frozen	not frozen
DF R full/full L0	not frozen	not frozen	not frozen
DF R full/full L1	not frozen	not frozen	not frozen
DF R full/euler L0	not frozen	not frozen	not frozen
DF R full/euler L1	not frozen	not frozen	not frozen
DF R identity/full L0	not frozen	not frozen	not frozen
DF R identity/full L1	not frozen	not frozen	not frozen
DF R identity/euler L0	not frozen	not frozen	not frozen
DF R identity/euler L1	not frozen	not frozen	not frozen
DF S full/full L0	not frozen	not frozen	not frozen
DF S full/full L1	not frozen	not frozen	not frozen
DF S full/euler L0	not frozen	not frozen	not frozen
DF S full/euler L1	not frozen	not frozen	not frozen
DF S identity/full L0	not frozen	not frozen	not frozen
DF S identity/full L1	not frozen	not frozen	not frozen
DF S identity/euler L0	not frozen	not frozen	not frozen
DF S identity/euler L1	not frozen	not frozen	not frozen
DF avgQ R full/full L0	not frozen	not frozen	not frozen
DF avgQ R full/full L1	not frozen	not frozen	not frozen
DF avgQ S full/full L0	not frozen	not frozen	not frozen
DF avgQ S full/full L1	not frozen	not frozen	not frozen
DF MH full/full	not frozen	not frozen	not frozen
DF avgQ MH full/full	not frozen	not frozen	not frozen
QGF avg-Q	not frozen	not frozen	not frozen
QGF smooth-J	not frozen	not frozen	not frozen
IPG	not frozen	not frozen	not frozen
IPG-CV	not frozen	not frozen	not frozen
IPG-RW	not frozen	not frozen	not frozen
FK	not frozen	not frozen	not frozen

not frozen means selection incomplete or task not started, not a failure score. These settings are chosen on validation only. Per-model test outcomes and all primary/positive/control configurations are separate CSV records.

Validation-only diagnosis: what repeated across FULL models?

Independent saved-artifact reports preserve all 29 selected results and exact settings for FULL seeds 11/23/37. Respectively 24/29, 6/29 and 21/29 conditions reached 8/8. These are tuning outcomes on shared starts, not heldout algorithm wins.

Seed 11: 29/29 FULL/HOLE primary validation success counts match. HOLE QGF=8/8 at its own kappa 40. Own settings can differ; these shared tuning starts can be ceiling-limited. The critic data hole guarantees neither Q error nor QGF failure.

Seed 23: 14/29 FULL/HOLE primary validation success counts match. HOLE QGF=8/8 at its own kappa 40. Own settings can differ; these shared tuning starts can be ceiling-limited. The critic data hole guarantees neither Q error nor QGF failure.

Seed 37: 25/29 FULL/HOLE primary validation success counts match. HOLE QGF=8/8 at its own kappa 40. Own settings can differ; these shared tuning starts can be ceiling-limited. The critic data hole guarantees neither Q error nor QGF failure.

Selected condition	s11 /8	s23 /8	s37 /8
Regular full-J/Euler L0	0	0	0
Regular full-J/Euler L1	0	0	0
Stochastic full-J/Euler L1	0	0	0
Stochastic full-J/Euler L0	7	6	6
IPG	0	0	1
IPG-CV	8	1	5
IPG-RW	8	0	4
FK	8	2	4
Raw MH	8	4	8
Averaged MH	8	7	8

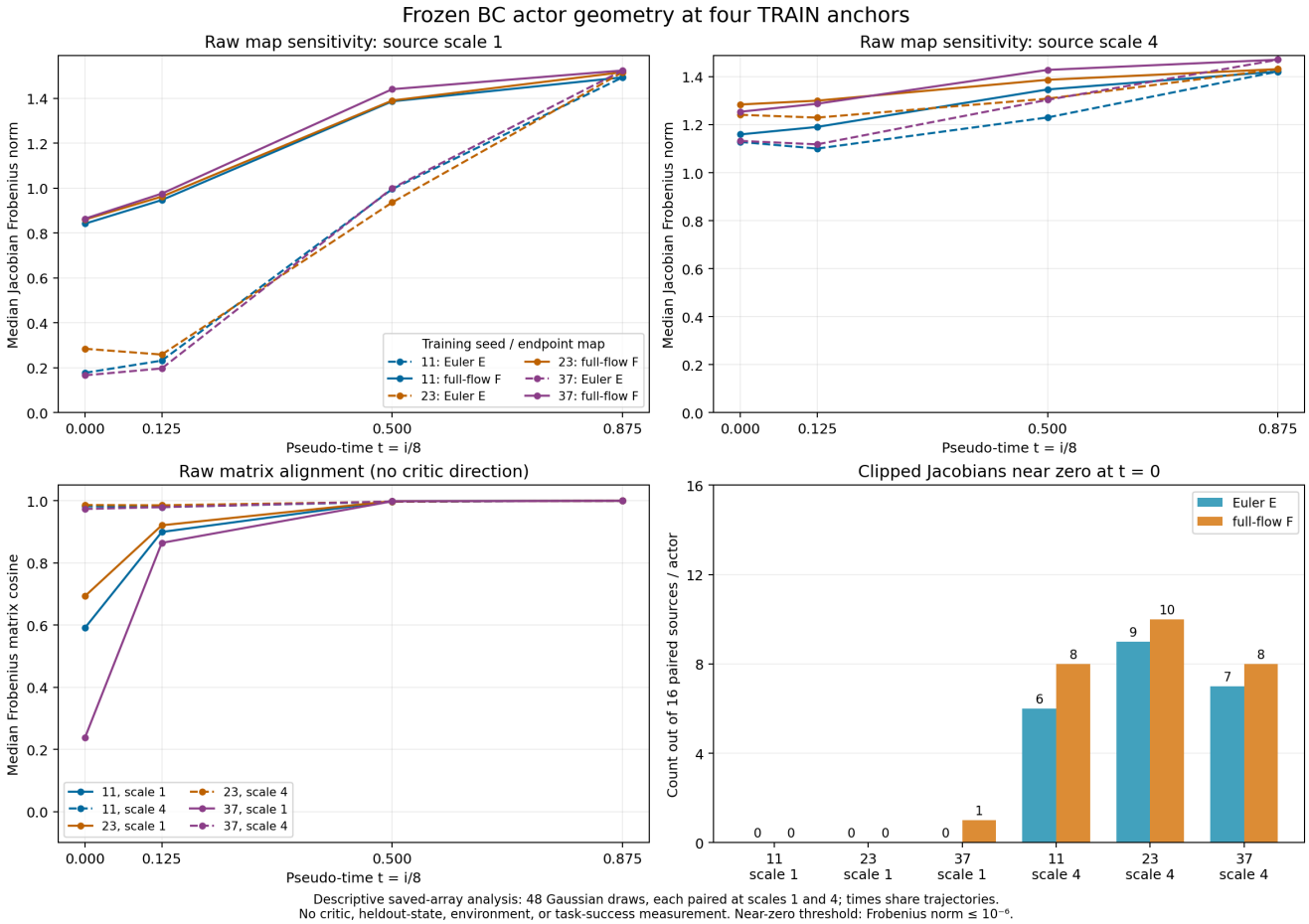
Seed 37 regular Euler L0 primary is BC fallback; its already-frozen positive .08/.1/I8 also 0/8. Small first-action guidance was seen in seed 11 and that secondary, but failed seed 23 Euler actions shifted 1.458/1.223 L2 with predicted raw-Q gains +.400/+.289. Seed 37 failed L1 Euler shifts 1.114/1.446 with gains +.229/+.450. Thus weak first guidance is not the common demonstrated cause. The ideal cancellation identity does not by itself diagnose learned closed-loop failures.

Ordinary IPG remained weak, even where ignored-weight ESS was high. Seed 37 IPG min/median ignored ESS 3.457/15.266 and max solve error .03809 reveal occasional concentration/numerical residual, but earlier failures occurred without that signature. RW shows occasional weight/lineage loss; FK final ESS 16 after resampling conceals pre-resampling minima near 1.3. Raw derivative maxima near 6700 are not centered update magnitudes.

Seed 37 raw/averaged MH have same k80/e.01/I256, 8/8 each, mean acceptance .853/.864 and source displacement 2.479/2.471. Other seeds use different eta/targets; acceptance is not mixing or Q accuracy. Source movement and initial-Q gains are descriptive. Both actor and critic change with trained seed; no causal separation or Q calibration follows. Actor-Jacobian optional status at snapshot: COMPLETED.

Actor geometry: Euler sensitivity and endpoint clipping

Actor-geometry check (TRAIN inputs only): 48 Gaussian draws from three frozen BC actors were paired at scales 1 and 4. At scale 1 and t=0, median raw Euler/full-flow Jacobian norm ratios were 0.218/0.330/0.205 for seeds 11/23/37. At scale 4 they were 0.968/0.967/0.861, while final clipping made 6/9/7 Euler and 8/10/8 full-flow Jacobians near zero out of 16 per actor. The t=7/8 maps and derivatives agree exactly because one step remains, not because their Jacobians are identity. All 192 predeclared FD rows are retained; no applicable guard failed (maximum normalized error 3.58e-4). This supports a location-dependent geometry mechanism, not a task-failure explanation: no Q direction or selected-loop state was evaluated, and seed 23 had large realized initial shifts in the separate validation diagnosis.



Raw and clipped Jacobians are distinct measurements. At later pseudo-times the derivative is with respect to the CURRENT x_t at an unguided BC prefix, not the original source z. The same 48 draws share their scale and time measurements; 1,536 Jacobian rows are not independent episodes or additional trained seeds.

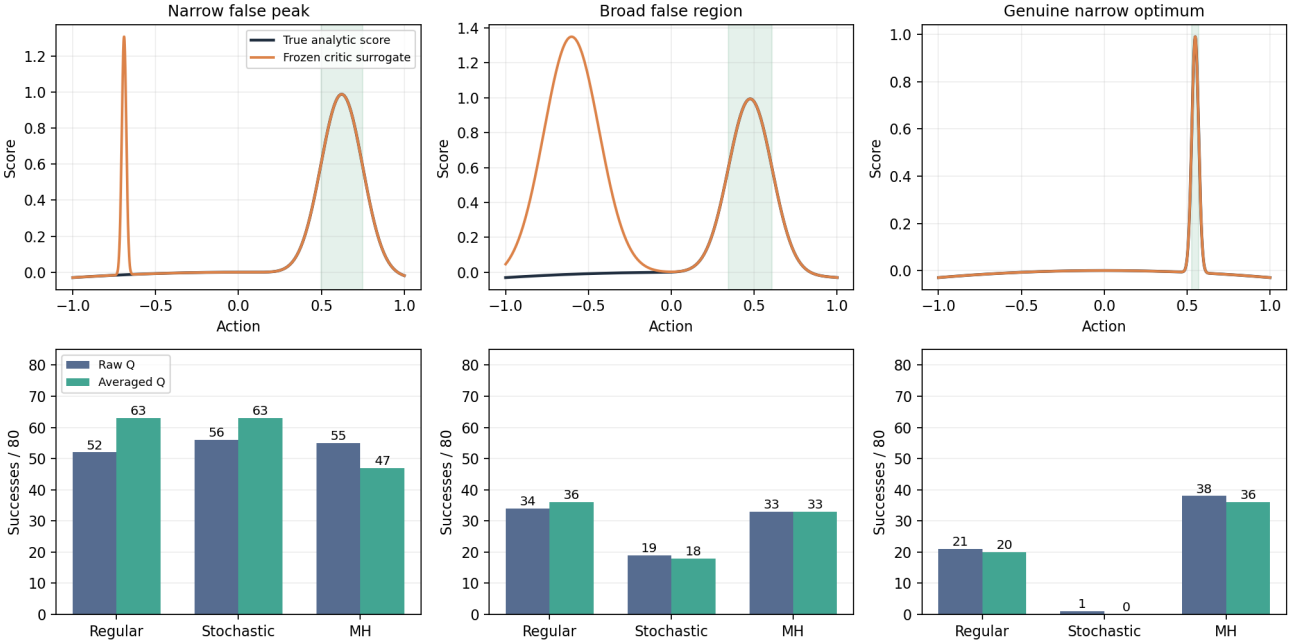
No sampled clip-region crossing or exact boundary occurred in the 192 finite-difference checks, so these passes do not empirically stress the cusp detector. Matrix alignment compares actor Jacobians, not critic-gradient directions. This descriptive TRAIN-input probe supplies neither a causal explanation for heldout failures nor evidence of critic repair.

Analytic diagnostic: narrow and broad critic errors

Constructed 1D final-point trials use known true scores plus injected critic errors. There are no next states, learned IQL models or physical robot episodes. Twelve cases include narrow false peaks, broad false regions and true narrow optima; validation settings freeze before tests.

Method	Narrow false /80	Broad false /80	True narrow /80
regular	52	34	21
stochastic	56	19	1
mh	55	33	38
qavg_regular	63	36	20
qavg_stochastic	63	18	0
qavg_mh	47	33	36

Controlled score-map diagnostic: averaging helps some cases and hurts others



Synthetic final-point trials, not IQL or robot episodes. Own validation settings; 4 cases × 20 test points per group. Top: first declared case in each group.

Q averaging helped regular/stochastic narrow-false cases, reduced MH there, and did not uniformly repair broad error. It can suppress true narrow optima. Actual unique test points: 1440. These are point outcomes, not trained seeds or physical successes. Own settings confound an isolated smoothing effect.

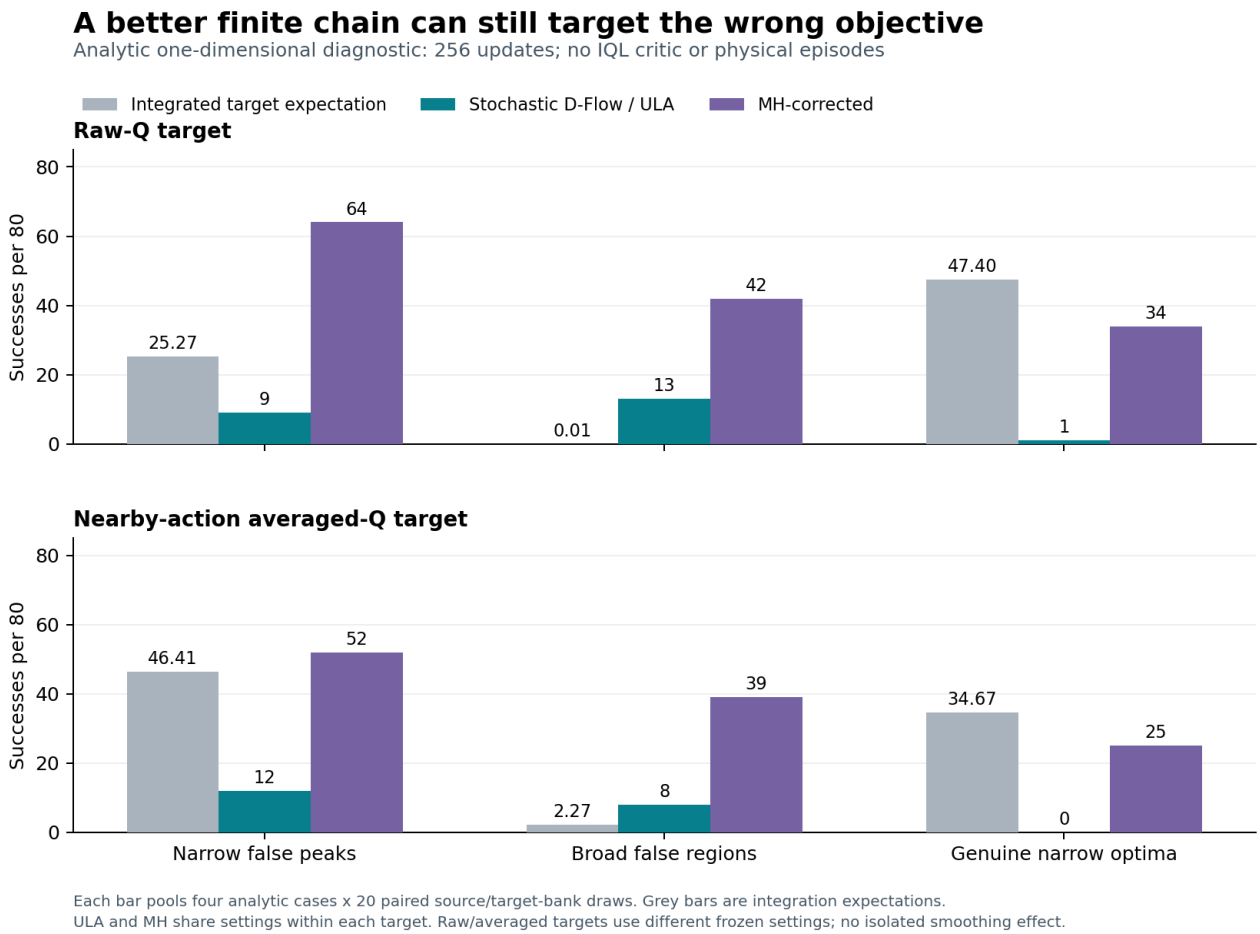
Matched targets: finite-chain success is not critic repair

Group/target	Target expected /80	ULA I256 /80	MH I256 /80
narrow false peak/raw	25.266	9	64
narrow false peak/qavg	46.410	12	52
broad false region/raw	0.015	13	42
broad false region/qavg	2.270	8	39
genuine narrow optimum/raw	47.405	1	34
genuine narrow optimum/qavg	34.667	0	25

Each target is held fixed from earlier validation selection. Fresh paired sources compare ULA/MH prefixes 1,4,16,64,256, not reselected settings. Broad-false raw target expectation is about .015/80, but finite MH gives 42/80: the finite chain has not equilibrated to the bad critic-defined target. Better finite outcomes can reflect protection from a misleading equilibrium, not repaired Q.

The study has 240 independent case/source random bundles, 960 paired algorithm chains and 4,800 correlated prefix endpoints. It contributes 0 physical episodes and uses 0 learned models. Five endpoints reuse each chain. Integration grid error is 1.07e-05; the tail-probability bound is 2.23e-33.

The true-narrow target expectation falls from 47.405/80 raw to 34.667/80 averaged. Targets differ and have their own frozen parameters, so no isolated causal averaging claim follows. Quadrature-guard failure and missing plotting dependency were preserved; finalization used saved arrays only.



Each bar pools four analytic cases x 20 paired source/target-bank draws. Grey bars are integration expectations.
ULA and MH share settings within each target. Raw/averaged targets use different frozen settings; no isolated smoothing effect.

Reproducibility, unsuccessful fixes and limits

The exact JSON snapshot binds every used config/cohort/source/checkpoint hash, selection, completed physical metadata/CSV/raw-trace hash and existing audit receipt. It is accompanied by all primary/positive settings, requested test rows, actual counts and file manifest CSVs. The source collector imports no numerical/model/environment library. Raw-array arithmetic and semantic aliases remain separately verified by the independent auditor.

Scientific runtime: CPython 3.11.11/macOS arm64, JAX/JAXlib 0.4.30, Flax 0.8.5, Optax 0.2.4, NumPy 1.26.4, MuJoCo 3.3.7, Gymnasium 0.29.1, OGBench 1.2.1. The inventory provides exact metadata/binary identities and separates the reporting runtime. Inference payload inputs fit about 40 MB before live results. The final manifest specifies whether the original training tables, including the inventoried 49.7 MB data extension, are included; the compiled runtime remains external. Current XML identities are recorded but do not retroactively create launch-time guards.

Preflight caught averaged-Q head-minimum/compiler interaction and batch-shape arithmetic sensitivity. An equivalent two-head elementwise minimum and fixed padded shape passed independent gradients/real-network/compiled checks before physical release. Original eight D-Flow/raw QGF remain unchanged. Numeric int/float identity duplication created redundant executions; they are counted and not presented as independent trials. Failed guards and source revisions remain preserved.

Operational actor/replay watchdog preparation found and fixed Darwin leader/session cleanup edge cases using only short stdlib dummy jobs. Optional actor diagnostic status: COMPLETED; exact replay status: REPLAY_PASS. Prepared does not mean executed. Replay, if completed, repeats the same 20 starts and adds 0 independent tests.

Historical launchers enforce this campaign deadline and completed caches. Full retraining additionally needs the inventoried actual next-state data and original partition/training sources. A small source/results bundle is not a portable offline runtime or guaranteed bitwise reproduction on another platform. Previous campaign/interim reports and frozen archives are preserved; final archive CRC/SHA verification belongs to the closure bundle receipt.

Separate post-deadline reproduction entry-point status: PREPARED_OR_PARTIAL. Preparation/static review does not establish compatibility or a successful replay. Its explicit release and new-output/runtime safeguards remain required. Prior robot manipulation and privileged-demo augmentation results stay in their immutable first-campaign reports and contribute no episodes to this campaign.

Bellman residuals with learned V measure self-consistency, not ground-truth nonterminal Q error. Missing coverage does not prove extrapolation failure; noise, smoothing and MH cannot identify true Q without information. No manipulation competence or general algorithm winner follows from these mazes/analytic points. Unequal search/inference budgets and only 3 trained seeds remain limitations.

Stress-task reset anchors originate from the offline TRAIN table used for learning. Sampler validation/test anchor episodes are disjoint, but the test is not on entirely new learning-state support. The 20 starts are paired across methods and trained seeds; only three training seeds are independent.

The first replay supervisor stopped before creating a child because this Mac's scientific Python 3.11 build lacks os.waitid. A reviewed Anaconda Python 3.13 supervisor override then ran the unchanged Python 3.11 scientific worker. This is an explicit command deviation from the prepared job spec. All 20 trajectories replayed exactly; the future reproduction wrapper remains unexecuted.

Companion compute tables cover completed physical-rollout batch wall/sampler timings only, counted once per actual configuration while retaining numeric rerun costs. Prescreen, training and other work are excluded from those tables; incomplete rollout timing remains unknown. No hardware-normalized Q/NFE query count was measured.

Use a fresh predeclared matched-actor/critic intervention or known-value sequential MDP to separate critic error from proxy-gradient and finite-sampling effects. Keep compute and target definitions matched and use new validation/test cohorts. Current heldout results must not become tuning data.